

Supplementary materials for “Structured prior distributions for the covariance matrix in latent factor models”

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Abstract

Supplementary material contained in this note include proofs and derivations of the results used in the paper, complete descriptions of the adaptive Gibbs samplers, further background and results on the simulation experiment and two applications.

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S1 Prior for the identified factor loadings matrix

Section 3 of the main manuscript describes two prior distributions for the unconstrained and unidentified factor loadings matrix $\mathbf{\Lambda} \in \mathbb{R}^{p \times k}$: (i) a matrix-normal prior, $\mathbf{\Lambda} \sim N_{p,k}(\mathbf{0}, \mathbf{\Phi}, \mathbf{\Psi})$ and (ii) a matrix- t prior, $\mathbf{\Lambda} \sim t_{p,k}(\varsigma, \mathbf{0}, \mathbf{\check{\Phi}}, \mathbf{\Psi})$. An identifiable factor loadings matrix $\tilde{\mathbf{\Lambda}} \in \mathbb{R}^{p \times k}$, which satisfies the PLT constraint, can be calculated from $\mathbf{\Lambda}$ by taking a LQ decomposition, $\mathbf{\Lambda} = \tilde{\mathbf{\Lambda}}\mathbf{Q}$, in which $\mathbf{Q}$ is a $k \times k$ orthogonal matrix. This decomposition will be unique as long as $p \geq k$ and $\mathbf{\Lambda}$ is full rank. Therefore, we can, in principle, calculate the prior density induced for the identifiable factor loadings matrix $\tilde{\mathbf{\Lambda}}$ by computing the joint density for $(\tilde{\mathbf{\Lambda}}, \mathbf{Q})$ and then integrating out $\mathbf{Q}$.

Slight modification to Theorem 2.1.13 in Muirhead [1] yields the Jacobian of the LQ decomposition from $\mathbf{\Lambda}$ to $(\tilde{\mathbf{\Lambda}}, \mathbf{Q})$ as

$$J(\mathbf{\Lambda} \rightarrow \tilde{\mathbf{\Lambda}}, \mathbf{Q}) = \prod_{i=1}^k \tilde{\lambda}_{ii}^{k-i} (\mathbf{Q} d\mathbf{Q}^T)$$

where $(\mathbf{Q} d\mathbf{Q}^T)$ is the Haar measure on the orthogonal group. Therefore, given a matrix normal or matrix t -density $p_{\Lambda}(\cdot)$ for the random matrix $\mathbf{\Lambda}$, the joint prior for $(\tilde{\mathbf{\Lambda}}, \mathbf{Q})$ has the form

$$p_{\Lambda}(\tilde{\mathbf{\Lambda}}\mathbf{Q}) \prod_{i=1}^k \tilde{\lambda}_{ii}^{k-i} (d\tilde{\mathbf{\Lambda}})(\mathbf{Q} d\mathbf{Q}^T). \quad (\text{S1})$$

Under certain conditions, a matrix normal distribution for $\mathbf{\Lambda}$ yields a prior for $\tilde{\mathbf{\Lambda}}$ that can be described using standard distributional families. The remainder of this section focuses on this specific special case.

If a random matrix $\mathbf{\Lambda}$ has a matrix normal distribution, $\mathbf{\Lambda} \sim N_{p,k}(\mathbf{0}, \mathbf{\Phi}, \mathbf{\Psi})$, then its probability density function is

$$p(\mathbf{\Lambda}) = \frac{1}{(2\pi)^{pk} |\mathbf{\Psi}|^{p/2} |\mathbf{\Phi}|^{k/2}} \exp \left\{ -\frac{1}{2} \text{tr} (\mathbf{\Phi}^{-1} \mathbf{\Lambda} \mathbf{\Psi}^{-1} \mathbf{\Lambda}^T) \right\}, \quad \mathbf{\Lambda} \in \mathbb{R}^{p \times k}.$$

Substituting this density into (S1) yields

$$\frac{1}{(2\pi)^{pk} |\mathbf{\Psi}|^{p/2} |\mathbf{\Phi}|^{k/2}} \exp \left\{ -\frac{1}{2} \text{tr} (\mathbf{\Phi}^{-1} \tilde{\mathbf{\Lambda}} \mathbf{Q} \mathbf{\Psi}^{-1} \mathbf{Q}^T \tilde{\mathbf{\Lambda}}^T) \right\} \prod_{i=1}^k \tilde{\lambda}_{ii}^{k-i} (d\tilde{\mathbf{\Lambda}})(\mathbf{Q} d\mathbf{Q}^T). \quad (\text{S2})$$

In the general case, $\tilde{\mathbf{\Lambda}}$ and $\mathbf{Q}$ are not independent *a priori*. Though the conditional distribution of $\mathbf{Q}$ given $\tilde{\mathbf{\Lambda}}$ is a matrix-variate Bingham distribution [2], its normalising constant is not computationally tractable and so the marginal distribution of $\tilde{\mathbf{\Lambda}}$ cannot, in general, be obtained in closed form. However, in the special case when $\mathbf{\Psi} = \psi \mathbf{I}_k$, the joint and marginal distributions of $\tilde{\mathbf{\Lambda}}$ and $\mathbf{Q}$ can be derived explicitly. To this end, let $\tilde{\lambda}_{(i)j} = (\tilde{\lambda}_{ij}, \dots, \tilde{\lambda}_{pj})^T$ denote the elements in rows i to p and column j of $\tilde{\mathbf{\Lambda}}$, and denote by $\Xi_{(i)}$ the submatrix of $\Xi = \mathbf{\Phi}^{-1}$ spanning rows i to p and columns i to p , partitioned as

$$\Xi_{(i)} = \begin{pmatrix} \xi_{ii} & \xi_{(i+1)i}^T \\ \xi_{(i+1)i} & \Xi_{(i+1)} \end{pmatrix} \quad (\text{S3})$$

where $\xi_{(i+1)i} = (\xi_{i+1,i}, \dots, \xi_{pi})^T$. We then have the following Theorem.

Theorem S1. *Let $\mathbf{\Lambda} = \tilde{\mathbf{\Lambda}}\mathbf{Q}$ be the LQ decomposition of the $p \times k$ random matrix $\mathbf{\Lambda} \sim N_{p,k}(\mathbf{0}, \mathbf{\Phi}, \psi \mathbf{I}_k)$ where $p \geq k$. Then the lower triangular matrix with positive diagonal elements $\tilde{\mathbf{\Lambda}}$ and the orthogonal matrix $\mathbf{Q}$ are independent. Moreover, the matrix $\mathbf{Q}$ is distributed according to the Haar measure over the orthogonal group while*

$\tilde{\mathbf{\Lambda}}$ is distributed such that $\tilde{\mathbf{\lambda}}_{(1)1}, \dots, \tilde{\mathbf{\lambda}}_{(k)k}$ are independent across columns with within-column distribution

$$\tilde{\lambda}_{ii}^2 \sim \text{Gam} \left\{ \frac{k-i+1}{2}, \frac{1}{2\psi} \left(\xi_{ii} - \boldsymbol{\xi}_{(i+1)i}^T \boldsymbol{\Xi}_{(i+1)}^{-1} \boldsymbol{\xi}_{(i+1)i} \right) \right\} \quad (\text{S4})$$

and

$$\tilde{\mathbf{\lambda}}_{(i+1)i} | \tilde{\lambda}_{ii} \sim \text{N}_{p-i} \left(-\tilde{\lambda}_{ii} \boldsymbol{\Xi}_{(i+1)}^{-1} \boldsymbol{\xi}_{(i+1)i}, \psi \boldsymbol{\Xi}_{(i+1)}^{-1} \right) \quad (\text{S5})$$

for column $i = 1, \dots, k$.

Proof. Substituting $\boldsymbol{\Psi} = \psi \mathbf{I}_k$ and $\boldsymbol{\Phi}^{-1} = \boldsymbol{\Xi}$ into (S2) and applying the cyclic property of the trace operator yields

$$\frac{1}{(2\pi)^{pk} |\boldsymbol{\Phi}|^{k/2} \psi^{pk/2}} \exp \left\{ -\frac{1}{2\psi} \text{tr} \left(\tilde{\mathbf{\Lambda}}^T \boldsymbol{\Xi} \tilde{\mathbf{\Lambda}} \right) \right\} \prod_{i=1}^k \tilde{\lambda}_{ii}^{k-i} (d\tilde{\mathbf{\Lambda}}) (\mathbf{Q} d\mathbf{Q}^T),$$

from which it is clear that $\tilde{\mathbf{\Lambda}}$ and $\mathbf{Q}$ are independent *a priori*, the distribution of $\mathbf{Q}$ is a (normalised) Haar measure, and the density of $\tilde{\mathbf{\Lambda}}$ is given by

$$p_{\tilde{\mathbf{\Lambda}}}(\tilde{\mathbf{\Lambda}}) \propto \exp \left\{ -\frac{1}{2\psi} \text{tr} \left(\tilde{\mathbf{\Lambda}}^T \boldsymbol{\Xi} \tilde{\mathbf{\Lambda}} \right) \right\} \prod_{i=1}^k \tilde{\lambda}_{ii}^{k-i}. \quad (\text{S6})$$

Noting that $\tilde{\mathbf{\lambda}}_{(1)i}^T = (\mathbf{0}^T, \tilde{\mathbf{\lambda}}_{(i)i}^T)$, the required trace in (S6) is clearly

$$\text{tr} \left(\tilde{\mathbf{\Lambda}}^T \boldsymbol{\Xi} \tilde{\mathbf{\Lambda}} \right) = \sum_{i=1}^k \tilde{\mathbf{\lambda}}_{(1)i}^T \boldsymbol{\Xi} \tilde{\mathbf{\lambda}}_{(1)i} = \sum_{i=1}^k \left(\mathbf{0}^T \tilde{\mathbf{\lambda}}_{(i)i}^T \right) \begin{pmatrix} * & * \\ * & \boldsymbol{\Xi}_{(i)} \end{pmatrix} \begin{pmatrix} \mathbf{0} \\ \tilde{\mathbf{\lambda}}_{(i)i} \end{pmatrix} = \sum_{i=1}^k \tilde{\mathbf{\lambda}}_{(i)i}^T \boldsymbol{\Xi}_{(i)} \tilde{\mathbf{\lambda}}_{(i)i}.$$

It follows that (S6) can be written as

$$p_{\tilde{\mathbf{\Lambda}}}(\tilde{\mathbf{\Lambda}}) \propto \prod_{i=1}^k \left\{ \exp \left(-\frac{1}{2\psi} \tilde{\mathbf{\lambda}}_{(i)i}^T \boldsymbol{\Xi}_{(i)} \tilde{\mathbf{\lambda}}_{(i)i} \right) \tilde{\lambda}_{ii}^{k-i} \right\}.$$

The factorisation over the columns of $\tilde{\mathbf{\Lambda}}$ implies immediately that $\tilde{\mathbf{\lambda}}_{(1)1}, \dots, \tilde{\mathbf{\lambda}}_{(k)k}$ are independent. Partitioning $\boldsymbol{\Xi}_{(i)}$ as in (S3), the density of $\tilde{\mathbf{\lambda}}_{(i)i}^T = (\tilde{\lambda}_{ii}, \tilde{\mathbf{\lambda}}_{(i+1)i}^T)$ is then proportional to

$$\begin{aligned} p(\tilde{\mathbf{\lambda}}_{(i)i}) &\propto \exp \left(-\frac{1}{2\psi} \tilde{\mathbf{\lambda}}_{(i)i}^T \boldsymbol{\Xi}_{(i)} \tilde{\mathbf{\lambda}}_{(i)i} \right) \tilde{\lambda}_{ii}^{k-i} \\ &= \exp \left\{ -\frac{1}{2\psi} \left(\tilde{\mathbf{\lambda}}_{(i+1)i} + \tilde{\lambda}_{ii} \boldsymbol{\Xi}_{(i+1)}^{-1} \boldsymbol{\xi}_{(i+1)i} \right)^T \boldsymbol{\Xi}_{(i+1)} \left(\tilde{\mathbf{\lambda}}_{(i+1)i} + \tilde{\lambda}_{ii} \boldsymbol{\Xi}_{(i+1)}^{-1} \boldsymbol{\xi}_{(i+1)i} \right) \right\} \\ &\quad \times \exp \left\{ -\frac{1}{2\psi} \left(\xi_{ii} - \boldsymbol{\xi}_{(i+1)i}^T \boldsymbol{\Xi}_{(i+1)}^{-1} \boldsymbol{\xi}_{(i+1)i} \right) \tilde{\lambda}_{ii}^2 \right\} \tilde{\lambda}_{ii}^{k-i}. \end{aligned} \quad (\text{S7})$$

We now need to introduce a second change of variables

$$\tilde{\lambda}_{ii} = \tilde{\lambda}_i^{*1/2}$$

so that

$$\frac{d\tilde{\lambda}_{ii}}{d\tilde{\lambda}_i^*} = 2^{-1} \tilde{\lambda}_i^{*-1/2}$$

and then (S7) becomes

$$\begin{aligned} p(\tilde{\lambda}_i^*, \tilde{\lambda}_{(i+1)i}) \\ \propto \exp \left\{ -\frac{1}{2\psi} \left(\tilde{\lambda}_{(i+1)i} + \tilde{\lambda}_i^{*1/2} \Xi_{(i+1)}^{-1} \boldsymbol{\xi}_{(i+1)i} \right)^\top \Xi_{(i+1)} \left(\tilde{\lambda}_{(i+1)i} + \tilde{\lambda}_i^{*1/2} \Xi_{(i+1)}^{-1} \boldsymbol{\xi}_{(i+1)i} \right) \right\} \\ \times \exp \left\{ -\frac{1}{2\psi} \left(\xi_{ii} - \boldsymbol{\xi}_{(i+1)i}^\top \Xi_{(i+1)}^{-1} \boldsymbol{\xi}_{(i+1)i} \right) \tilde{\lambda}_i^* \right\} \tilde{\lambda}_i^{*(k-i-1)/2}. \end{aligned}$$

It follows that

$$\tilde{\lambda}_i^* \equiv \tilde{\lambda}_{ii}^2 \sim \text{Gam} \left(\frac{k-i+1}{2}, \frac{1}{2\psi} \left(\xi_{ii} - \boldsymbol{\xi}_{(i+1)i}^\top \Xi_{(i+1)}^{-1} \boldsymbol{\xi}_{(i+1)i} \right) \right),$$

and

$$\tilde{\lambda}_{(i+1)i} | \tilde{\lambda}_i^* \sim \text{N}_{p-i} \left(-\tilde{\lambda}_i^{*1/2} \Xi_{(i+1)}^{-1} \boldsymbol{\xi}_{(i+1)i}, \psi \Xi_{(i+1)}^{-1} \right),$$

for $i = 1, \dots, k$ as required. $\square$

In the special case when $\boldsymbol{\Phi} = \mathbf{I}_p$ in addition to $\boldsymbol{\Psi} = \psi \mathbf{I}_k$, we obtain the following corollary to Theorem S1, which is equivalent to Theorem 2.1 in Leung and Drton [3].

Corollary S1. *Let $\boldsymbol{\Lambda} = \tilde{\boldsymbol{\Lambda}} \mathbf{Q}$ be the LQ decomposition of the $p \times k$ random matrix $\boldsymbol{\Lambda} \sim \text{N}_{p,k}(\mathbf{0}, \mathbf{I}_p, \psi \mathbf{I}_k)$ where $p \geq k$. Then the lower triangular matrix with positive diagonal elements $\tilde{\boldsymbol{\Lambda}}$ and the orthogonal matrix $\mathbf{Q}$ are independent, the matrix $\mathbf{Q}$ is distributed according to the Haar measure over the orthogonal group, while the non-zero elements in $\tilde{\boldsymbol{\Lambda}}$ are independent and, for $i = 1, \dots, k$, distributed such that*

$$\tilde{\lambda}_{ii}^2 \sim \text{Gam} \left(\frac{k-i+1}{2}, \frac{1}{2\psi} \right)$$

and

$$\tilde{\lambda}_{ij} \sim \text{N}(0, \psi), \quad j = 1, \dots, i-1.$$

Proof. Set $\boldsymbol{\Xi} = \mathbf{I}_p$ and then $\boldsymbol{\Xi}_{(i)} = \mathbf{I}_{p-i+1}$, $\xi_{ii} = 1$ and $\boldsymbol{\xi}_{(i+1)i} = \mathbf{0}$. Substituting these expressions into (S4) and (S5) yields the desired result. $\square$

S2 Structured prior distributions

S2.1 Proofs of results concerning the constrained prior expectation

This section makes extensive use of the results from Chapter 17 of Seber [4], as detailed below. We begin by proving Proposition 1 and Theorem 1 of the main manuscript.

S2.1.1 On Proposition 1

Proof of Proposition 1. The squared Frobenius distance between $\mathbf{\Lambda}\mathbf{\Lambda}^\top$ and $\mathbf{D}$, denoted $d^2 = d\{\mathbf{\Lambda}\mathbf{\Lambda}^\top, \mathbf{D}\}^2$, is given by

$$\begin{aligned} d^2 &= \text{tr} \{ (\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{D})^\top (\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{D}) \} \\ &= \text{tr} \{ (\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{D})^2 \}. \end{aligned} \quad (\text{S8})$$

Differentiating with respect to $\mathbf{\Lambda}$ using Results 17.38 and 17.40 of Seber [4] yields:

$$\frac{\partial d^2}{\partial \mathbf{\Lambda}} = 4\mathbf{\Lambda}^\top (\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{D}) \quad (\text{S9})$$

in which $\partial d^2 / \partial \mathbf{\Lambda}$ is a $k \times p$ matrix whose (i, j) -th entry is $\partial d^2 / \partial \lambda_{ji}$. Setting the matrix of derivatives equal to zero and taking the transpose therefore implies

$$(\mathbf{\Lambda}\mathbf{\Lambda}^\top - \mathbf{D})\mathbf{\Lambda} = \mathbf{0}. \quad (\text{S10})$$

Denote by $\mathcal{E}$ a k -dimensional invariant subspace of $\mathbf{D}$. Now choose a linear map $\mathbf{P} : \mathbb{R}^p \rightarrow \mathbb{R}^k$ such that $\mathbf{P}\mathbf{P}^\top$ is the identity and $\mathcal{E}$ is the orthogonal complement of the null space of $\mathbf{P}^\top \mathbf{P}$. Note that $\mathbf{P}^\top \mathbf{P}$ is an orthogonal projection onto $\mathcal{E}$.

Substituting $\mathbf{\Lambda}^\top = \mathbf{P}\mathbf{D}^{1/2}$ into the left-hand-side of (S10) yields

$$\begin{aligned} \mathbf{\Lambda}\mathbf{\Lambda}^\top \mathbf{\Lambda} - \mathbf{D}\mathbf{\Lambda} &= \mathbf{D}^{1/2} \mathbf{P}^\top \mathbf{P} \mathbf{D} \mathbf{P}^\top - \mathbf{D} \mathbf{D}^{1/2} \mathbf{P}^\top \\ &= \mathbf{D}^{3/2} \mathbf{P}^\top - \mathbf{D}^{3/2} \mathbf{P}^\top \\ &= \mathbf{0} \end{aligned}$$

where $\mathbf{P}^\top \mathbf{P} \mathbf{D} \mathbf{P}^\top = \mathbf{D} \mathbf{P}^\top$ because $\mathbf{D} \mathbf{P}^\top$ lies in $\mathcal{E}$ and so the projection $\mathbf{P}^\top \mathbf{P}$ acts as the identity. It follows that $\mathbf{\Lambda} = \mathbf{D}^{1/2} \mathbf{P}^\top$ is a critical point.

The value of the squared Frobenius distance (S8) at this point is given by

$$\begin{aligned} d^2 &= \text{tr} \left\{ \left(\mathbf{D}^{1/2} \mathbf{P}^\top \mathbf{P} \mathbf{D}^{1/2} - \mathbf{D} \right)^2 \right\} \\ &= \text{tr} \left[\{ (\mathbf{I}_p - \mathbf{P}^\top \mathbf{P}) \mathbf{D} \}^2 \right]. \end{aligned}$$

Since $\mathbf{I}_p - \mathbf{P}^\top \mathbf{P}$ is a projection onto the orthogonal complement of $\mathcal{E}$, d^2 is therefore equal to the sum of the squares of the eigenvalues of the $p - k$ eigenvectors of $\mathbf{D}$ that do not correspond to $\mathcal{E}$.

Order the p eigenvectors $\{v_i\}_{i \in [1..p]}$ in such a way that their eigenvalues $\{m_i\}_{i \in [1..p]}$ satisfy $m_i \geq m_j$ if $i < j$. Define $\mathcal{E} = \text{span}\{\{v_i\}_{i \in [1..k]}\}$. Clearly such a choice of $\mathcal{E}$ will have the minimal value of $d^2 = \sum_{i=k+1}^p m_i^2$.

If $m_k \neq m_{k+1}$, then $\mathcal{E}$ is unique. There is nevertheless a degeneracy in the set of critical points, due to the invariance of the distance to the action of $O(k)$. If we replace $\mathbf{P}$ by $\mathbf{O}\mathbf{P}$, where $\mathbf{O} \in O(k)$, then the properties of $\mathbf{P}$ are still satisfied, and $\mathbf{\Lambda}^\top = \mathbf{O}\mathbf{P}\mathbf{D}^{1/2}$ is also a critical point. The set of critical points is therefore $k(k-1)/2$ -dimensional. Note that all these produce the same value of $\mathbf{\Lambda}\mathbf{\Lambda}^\top$, which is therefore still unique.

Since one possible value of $\mathbf{P}$ is clearly $\mathbf{U}^{(k)}$, the matrix whose columns are the $\{v_i\}_{i \in [1..k]}$, the product $\mathbf{\Lambda}\mathbf{\Lambda}^\top$ that minimises d^2 is

$$\mathbf{\Lambda}\mathbf{\Lambda}^\top = \mathbf{D}^{1/2} \mathbf{U}^{(k)} \mathbf{U}^{(k)\top} \mathbf{D}^{1/2}.$$

□

The above assumed that $m_k \neq m_{k+1}$. If this is false, then there is a further degeneracy present, this time in the choice of the subspace $\mathcal{E}$, as follows.

In general, it will be the case that $m_{i^*+1} = m_{j^*}$ for some $i^* < k$ and $j^* \geq k$ such that $m_{i^*} > m_{i^*+1}$ and $m_{j^*} > m_{j^*+1}$. There is then a q -dimensional eigenspace $\mathcal{V}$ spanned by the eigenvectors corresponding to the eigenvalue m_{i^*+1} . Let $\tilde{\mathcal{E}}$ be the space spanned by $\{v_i\}_{i \in [1..i^*]}$. Then the spaces $\mathcal{E}_{\mathcal{U}} = \tilde{\mathcal{E}} \oplus \mathcal{U}$, where $\mathcal{U}$ is any $(k - i^*)$ -dimensional subspace of $\mathcal{V}$, will all have the same value of d^2 , and this will be the minimal value. If $m_k > m_{k+1}$, then $q = 1$, and the subspace $\mathcal{U}$ will be unique; this is the case we have already discussed. If $m_k = m_{k+1}$, then $q > 1$, and there will be a $(k - i^*)(j^* - k)$ -dimensional space of possible critical points corresponding to different choices of $\mathcal{U}$, and hence of $\mathcal{E}$. These different choices of $\mathcal{E}$ do correspond to different values of the product $\mathbf{\Lambda}\mathbf{\Lambda}^\top$, because the image of $\mathbf{\Lambda}\mathbf{\Lambda}^\top$ is $\mathcal{E}$.

The maps $\mathbf{\Lambda}_{\mathcal{O}\mathcal{U}}^\top = \mathbf{O}\mathbf{P}_{\mathcal{U}}\mathbf{D}^{1/2}$, where $\mathbf{P}_{\mathcal{U}}$ is one of the linear maps defined by $\mathcal{E}_{\mathcal{U}}$, are therefore all critical points corresponding to the same minimal value of distance. Taking into account the previous degeneracy, the full space of critical points thus has $k(k-1)/2 + (k - i^*)(j^* - k)$ dimensions. The corresponding space of products $\mathbf{\Lambda}\mathbf{\Lambda}^\top$ is however only $(k - i^*)(j^* - k)$ -dimensional.

To show that these minimizing critical points are all local minima (necessarily non-strict due to degeneracy), we compute the Hessian matrix of the squared Frobenius distance d^2 . The Hessian can be defined as

$$\nabla^2 d^2 = \frac{\partial d^2}{\partial \text{vec}(\mathbf{\Lambda}) \partial \text{vec}(\mathbf{\Lambda})^\top} = \frac{\partial}{\partial \text{vec}(\mathbf{\Lambda})} \left\{ \frac{\partial d^2}{\partial \text{vec}(\mathbf{\Lambda})^\top} \right\} = \left\{ \frac{\partial}{\partial \text{vec}(\mathbf{\Lambda})^\top} \text{vec} \left(\frac{\partial d^2}{\partial \mathbf{\Lambda}} \right) \right\}^\top.$$

Substituting in the expression for the first derivative from (S9) yields

$$(\nabla^2 d^2)^\top = 4 \frac{\partial}{\partial \text{vec}(\mathbf{\Lambda})^\top} \text{vec}(\mathbf{\Lambda} \mathbf{\Lambda}^\top \mathbf{\Lambda}) - 4 \frac{\partial}{\partial \text{vec}(\mathbf{\Lambda})^\top} \text{vec}(\mathbf{D} \mathbf{\Lambda})$$

in which Result 17.30(b) from Seber [4] gives

$$\frac{\partial}{\partial \text{vec}(\mathbf{\Lambda})^\top} \text{vec}(\mathbf{D} \mathbf{\Lambda}) = \mathbf{I}_k \otimes \mathbf{D}$$

and two applications of the product rule (Result 17.30(h)), along with Results 17.30(a) and 17.30(c) from Seber [4] yields (dropping the vec notation for ease of reading):

$$\begin{aligned} \frac{\partial}{\partial \mathbf{\Lambda}^\top} (\mathbf{\Lambda} \mathbf{\Lambda}^\top \mathbf{\Lambda}) &= (\mathbf{\Lambda} \otimes \mathbf{I}_p)^\top \frac{\partial}{\partial \mathbf{\Lambda}^\top} (\mathbf{\Lambda} \mathbf{\Lambda}^\top) + (\mathbf{I}_k \otimes \mathbf{\Lambda} \mathbf{\Lambda}^\top) \frac{\partial \mathbf{\Lambda}}{\partial \mathbf{\Lambda}^\top} \\ &= (\mathbf{\Lambda} \otimes \mathbf{I}_p)^\top \left\{ (\mathbf{\Lambda}^\top \otimes \mathbf{I}_p)^\top \frac{\partial \mathbf{\Lambda}}{\partial \mathbf{\Lambda}^\top} + (\mathbf{I}_p \otimes \mathbf{\Lambda}) \frac{\partial \mathbf{\Lambda}^\top}{\partial \mathbf{\Lambda}^\top} \right\} \\ &\quad + (\mathbf{I}_k \otimes \mathbf{\Lambda} \mathbf{\Lambda}^\top) \\ &= (\mathbf{\Lambda} \otimes \mathbf{I}_p)^\top \{ (\mathbf{\Lambda}^\top \otimes \mathbf{I}_p)^\top + (\mathbf{I}_p \otimes \mathbf{\Lambda}) \mathbf{T} \} + (\mathbf{I}_k \otimes \mathbf{\Lambda} \mathbf{\Lambda}^\top) \end{aligned}$$

where $\mathbf{T} : \mathbb{R}^k \otimes \mathbb{R}^p \rightarrow \mathbb{R}^p \otimes \mathbb{R}^k$ is the transposition operator. Therefore

$$(\nabla^2 d^2)^T = 4 \{ (\mathbf{\Lambda}^\top \mathbf{\Lambda} \otimes \mathbf{I}_p) + (\mathbf{I}_k \otimes \mathbf{\Lambda} \mathbf{\Lambda}^\top) + (\mathbf{\Lambda}^\top \otimes \mathbf{\Lambda}) \mathbf{T} - (\mathbf{I}_k \otimes \mathbf{D}) \} .$$

We now need to find the eigenvalues of this Hessian. Using the above expression, the eigenvalue equation for the action of the second derivative at $\mathbf{\Lambda}$ on a tangent vector $\delta \mathbf{\Lambda}$ is given by:

$$4(\delta \mathbf{\Lambda} \mathbf{\Lambda}^\top \mathbf{\Lambda} + \mathbf{\Lambda} \mathbf{\Lambda}^\top \delta \mathbf{\Lambda} + \mathbf{\Lambda} \delta \mathbf{\Lambda}^\top \mathbf{\Lambda} - \mathbf{D} \delta \mathbf{\Lambda}) = \mu \delta \mathbf{\Lambda} .$$

We are interested in the behaviour at a critical point. Let us fix $\mathcal{U} \subset \mathcal{V}$, and hence $\mathcal{E}$. There remains a set of critical points isomorphic to $O(k)$; let us pick an arbitrary representative, leading to a specific critical point $\mathbf{\Lambda}^\top = \mathbf{P} \mathbf{D}^{1/2}$. Let us also pick an orthonormal basis $\{v_i\}_{i \in [1..k]}$ for $\mathcal{E}$ and for its orthogonal complement $\{v_i\}_{i \in [(k+1)..p]}$ consisting of eigenvectors of $\mathbf{D}$, ordered as before in order of decreasing eigenvalues.

Define $e_a = \mathbf{P} v_a$ for $i \in [1..k]$. The $\{e_a\}$ form an orthonormal basis for $\mathbb{R}^k$. The operator $\mathbf{P}$ can then be written in the form $\mathbf{P} = \sum_{a \in k} e_a \otimes v_a^\top$. We will henceforth use letters at the beginning of the alphabet as subscripts taking values in $[1..k]$.

Now define $v_{ia} = v_i \otimes e_a^\top$, with $a \in [1..k]$, $i \in [1..p]$. Note that these quantities are in $\mathbb{R}^{p \times k}$, the tangent space at $\mathbf{\Lambda}$, as required.

It can then be verified by direct calculation that the eigenvalues and eigenvectors of the above action are as follows.

- There are $k(k+1)/2$ eigenvectors of the form $m_a^{1/2} v_{ab} + m_b^{1/2} v_{ba}$, with eigenvalues of the form $m_a + m_b$. These eigenvalues are clearly positive.

- There are $k(p - k)$ eigenvectors of the form v_{ja} , where $v_j \in \mathcal{E}^\perp$, with eigenvalues of the form $m_a - m_j$. These eigenvalues are all non-negative. They are zero if both v_a and v_j are in $\mathcal{V}$, but otherwise are positive. There are $(k - i^*)(j^* - k)$ such zero eigenvalues. These are to be expected, and simply correspond to changes of the subspace U . Note that if m_k is not degenerate, then all such eigenvalues are positive.
- There are $k(k - 1)/2$ eigenvectors of the form $m_b^{1/2}v_{ba} - m_a^{1/2}v_{ab}$, with zero eigenvalues. These correspond to the degeneracy produced by the invariance of the distance to the action of $O(k)$.

Note that the total number of eigenvectors is (correctly) kp , the dimension of the tangent space at each point of $\mathbb{R}^{k \times p}$. Thus we see that all eigenvalues of the Hessian are non-negative, with the zero eigenvalues corresponding either to the expected degeneracy due to the action of $O(k)$, or to degeneracy caused by degenerate eigenvalues of D , as outlined earlier.

S2.1.2 On Theorem 1

Proof of Theorem 1. We defined the constrained expectation of Δ as $E_F(\Delta) = L_0 L_0^\top$ where

$$\begin{aligned} L_0 &= \min_{L \in \mathbb{R}^{p \times k}} E_\Lambda \{d(\Lambda \Lambda^\top, L L^\top)^2\} \\ &= \min_{L \in \mathbb{R}^{p \times k}} E_\Lambda \left[\text{tr} \left\{ (\Lambda \Lambda^\top - L L^\top)^2 \right\} \right]. \end{aligned}$$

Differentiating under the integral sign gives the condition

$$E_\Lambda \{ (L L^\top - \Lambda \Lambda^\top) L \} = \mathbf{0}$$

which reduces to

$$\{L L^\top - E_\Lambda(\Delta)\} L = \mathbf{0}.$$

The result then follows immediately from the proof of Proposition 1. $\square$

Although the rotational invariance of factor models is widely understood, it is worth emphasising the lack of uniqueness in constrained expectation caused by repeated eigenvalues in $E(\Delta)$. This is because many prior distributions in the literature construct a prior for Δ whose expectation is a scaled identity matrix with only one distinct eigenvalue; see, for example, Leung and Drton [3] and Bhattacharya and Dunson [5]. In applications where the data provide little information to distinguish between the different constrained expectations, this is likely to manifest as a lack of identification of Δ , which will contribute to poor mixing of MCMC samplers.

S2.2 Prior moments of the shared variation matrix

The (unconstrained) expected value of the shared variation matrix $\Delta = \Lambda \Lambda^\top$ is a $p \times p$ matrix defined by $E(\Delta) = \{E(\delta_{ij})\}$ and, in this case, $E(\Delta) = E(\Lambda \Lambda^\top)$ which is clearly symmetric.

Define the $p^2 \times p^2$ block matrix $\mathbf{V}$ by

$$\mathbf{V} = \text{Var}\{\text{vec}(\Delta^T)\} = \begin{pmatrix} \text{Var}(\delta_1) & \text{Cov}(\delta_1, \delta_2) & \cdots & \text{Cov}(\delta_1, \delta_n) \\ \text{Cov}(\delta_2, \delta_1) & \text{Var}(\delta_2) & \cdots & \text{Cov}(\delta_2, \delta_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\delta_n, \delta_1) & \text{Cov}(\delta_n, \delta_2) & \cdots & \text{Var}(\delta_n) \end{pmatrix}$$

where δ_i^T denotes the i^{th} row of $\Delta = \Lambda\Lambda^T$. There is clearly some redundancy in this matrix because Δ is itself symmetric. For example, $\text{Cov}(\delta_{11}, \delta_{12})$ appears in the upper triangle as both the $(1, 2)^{\text{th}}$ element in $\text{Var}(\delta_1)$ and the $(1, 1)^{\text{th}}$ element in $\text{Cov}(\delta_1, \delta_2)$. Denote by $\text{vecp}(\cdot)$ the operator that converts a symmetric matrix into a vector by stacking the on- and above-diagonal elements columnwise. The covariance matrix, $\text{Var}(\Delta)$, can therefore be interpreted as the $p(p+1)/2 \times p(p+1)/2$ matrix

$$\text{Var}(\Delta) = \text{Var}\{\text{vecp}(\Delta^T)\} = B_p^T \mathbf{V} B_p$$

in which $\text{vecp}(\Delta) = (\delta_{11}, \delta_{12}, \delta_{22}, \delta_{13}, \dots, \delta_{pp})^T = \text{vecp}(\Delta^T)$ and B_p is the $p^2 \times p(p+1)/2$ transition matrix whose j^{th} column comprises $p^2 - 1$ zeros and a one; the one appears in place i if the i^{th} element in $\text{vec}(\Delta^T)$ is the j^{th} element in $\text{vecp}(\Delta^T)$ and its operation on $\mathbf{V}$ is therefore to pick out the unique elements above the diagonal.

Consider the $p \times p$ matrix occupying the $(i, k)^{\text{th}}$ block of $\mathbf{V}$

$$\text{Cov}(\delta_i, \delta_k) = \text{Cov}(\Lambda\lambda_i, \Lambda\lambda_k) = E(\Lambda\Lambda^T \mathbf{e}_i \mathbf{e}_k^T \Lambda\Lambda^T) - E(\Lambda\Lambda^T \mathbf{e}_i) E(\Lambda\Lambda^T \mathbf{e}_k)^T$$

where λ_i^T denotes the i^{th} row of Λ and $\mathbf{e}_i$ denotes a p -vector of 0s with a 1 in the i^{th} place. Pre-multiplication by $\mathbf{e}_j^T$ and post-multiplication by $\mathbf{e}_\ell$ then picks out the $(i, \ell)^{\text{th}}$ element of the block, which can be written as

$$\text{Cov}(\delta_{ij}, \delta_{k\ell}) = \mathbf{e}_j^T E(\Lambda\Lambda^T \mathbf{e}_i \mathbf{e}_k^T \Lambda\Lambda^T) \mathbf{e}_\ell - \mathbf{e}_j^T E(\Lambda\Lambda^T) \mathbf{e}_i \mathbf{e}_k^T E(\Lambda\Lambda^T) \mathbf{e}_\ell \quad (\text{S11})$$

where $E(\Lambda\Lambda^T)^T = E(\Lambda\Lambda^T)$ because $\Lambda\Lambda^T$ is symmetric.

In order to compute the first and second order moments in a distribution for Δ we therefore need to compute the second order moment $E(\Lambda\Lambda^T)$ and the fourth order moment $E(\Lambda\Lambda^T \mathbf{e}_i \mathbf{e}_k^T \Lambda\Lambda^T)$ under the distribution for the random matrix Λ .

S2.3 Prior based on the matrix normal distribution

The calculations that follow make use of numerous numbered Theorems from Chapters 2, 3 and 4 of Gupta and Nagar [6] (hereafter GN), as indicated. Suppose that $\Lambda \sim N_{p,k}(\mathbf{0}, \Phi, \Psi)$. From GN Theorem 2.3.5(ii), it follows that the required expectation is given by

$$E(\Delta) = E(\Lambda\Lambda^T) = \text{tr}(\Psi)\Phi. \quad (\text{S12})$$

The general form for the covariance between δ_{ij} and $\delta_{k\ell}$ was given in (S11). For its calculation, we first need to compute $E(\Lambda\Lambda^T \mathbf{e}_i \mathbf{e}_k^T \Lambda\Lambda^T)$. This follows from GN

Theorem 2.3.8(v) as

$$E(\mathbf{\Lambda}\mathbf{\Lambda}^T \mathbf{e}_i \mathbf{e}_k^T \mathbf{\Lambda}\mathbf{\Lambda}^T) = \text{tr}(\mathbf{\Psi}^2) \phi_{ik} \mathbf{\Phi} + \text{tr}(\mathbf{\Psi})^2 \phi^{(i)} \phi_k^T + \text{tr}(\mathbf{\Psi}^2) \phi^{(k)} \phi_i^T$$

in which $\phi^{(i)}$ and ϕ_i^T denote the i^{th} column and row of $\mathbf{\Phi}$, respectively. Pre-multiplication by $\mathbf{e}_j^T$ and post-multiplication by $\mathbf{e}_\ell$ then yields

$$\mathbf{e}_j^T E(\mathbf{\Lambda}\mathbf{\Lambda}^T \mathbf{e}_i \mathbf{e}_k^T \mathbf{\Lambda}\mathbf{\Lambda}^T) \mathbf{e}_\ell = \text{tr}(\mathbf{\Psi}^2) \phi_{ik} \phi_{j\ell} + \text{tr}(\mathbf{\Psi})^2 \phi_{ij} \phi_{k\ell} + \text{tr}(\mathbf{\Psi}^2) \phi_{i\ell} \phi_{jk}.$$

Using (S12), the product of the second order moments in (S11) is given by

$$\mathbf{e}_j^T E(\mathbf{\Lambda}\mathbf{\Lambda}^T) \mathbf{e}_i \mathbf{e}_k^T E(\mathbf{\Lambda}\mathbf{\Lambda}^T) \mathbf{e}_\ell = \text{tr}(\mathbf{\Psi})^2 \phi_{ij} \phi_{k\ell}.$$

The covariance between δ_{ij} and $\delta_{k\ell}$ is therefore

$$\text{Cov}(\delta_{ij}, \delta_{k\ell}) = \text{tr}(\mathbf{\Psi}^2) (\phi_{ik} \phi_{j\ell} + \phi_{i\ell} \phi_{jk}).$$

In the special case where $i = k$ and $j = \ell$, the variance of δ_{ij} is given by

$$\text{Var}(\delta_{ij}) = \text{tr}(\mathbf{\Psi}^2) (\phi_{ii} \phi_{jj} + \phi_{ij}^2).$$

S2.4 Prior based on the matrix- t distribution

The calculations in this section rely on the representation of a matrix- t random variable expressed through (7) of the manuscript, from which it can be deduced that $\mathbf{\Lambda}|\mathbf{S} \sim \text{N}_{p,k}(\mathbf{M}, \mathbf{S}^{-1}, \mathbf{\Psi})$. Suppose that $\mathbf{\Lambda} \sim \text{t}_{p,k}(\varsigma, \mathbf{0}, \check{\mathbf{\Phi}}, \mathbf{\Psi})$. From GN Theorem 4.3.2(ii), it follows that the required expectation is given by

$$E(\mathbf{\Delta}) = E(\mathbf{\Lambda}\mathbf{\Lambda}^T) = \frac{\text{tr}(\mathbf{\Psi})}{\varsigma - 2} \check{\mathbf{\Phi}} \quad (\text{S13})$$

for $\varsigma > 2$.

In the calculation of the covariances, the first term in (S11) can be computed using the Law of Total Expectation according to which

$$\begin{aligned} E(\mathbf{\Lambda}\mathbf{\Lambda}^T \mathbf{e}_i \mathbf{e}_k^T \mathbf{\Lambda}\mathbf{\Lambda}^T) &= E_S \{ E_{\Lambda|S}(\mathbf{\Lambda}\mathbf{\Lambda}^T \mathbf{e}_i \mathbf{e}_k^T \mathbf{\Lambda}\mathbf{\Lambda}^T) \} \\ &= E_S \{ \text{tr}(\mathbf{\Psi}^2) \text{tr}(\mathbf{e}_i \mathbf{e}_k^T \mathbf{S}^{-1}) \mathbf{S}^{-1} + \text{tr}(\mathbf{\Psi})^2 \mathbf{S}^{-1} \mathbf{e}_i \mathbf{e}_k^T \mathbf{S}^{-1} + \text{tr}(\mathbf{\Psi}^2) \mathbf{S}^{-1} \mathbf{e}_k \mathbf{e}_i^T \mathbf{S}^{-1} \} \\ &= \text{tr}(\mathbf{\Psi}^2) E_S \{ \text{tr}(\mathbf{e}_i \mathbf{e}_k^T \mathbf{S}^{-1}) \mathbf{S}^{-1} \} + \text{tr}(\mathbf{\Psi})^2 E_S (\mathbf{S}^{-1} \mathbf{e}_i \mathbf{e}_k^T \mathbf{S}^{-1}) \\ &\quad + \text{tr}(\mathbf{\Psi}^2) E_S (\mathbf{S}^{-1} \mathbf{e}_k \mathbf{e}_i^T \mathbf{S}^{-1}). \end{aligned} \quad (\text{S14})$$

Now, if $\varsigma > 4$ we define

$$c_2 = \{(\varsigma - 1)(\varsigma - 2)(\varsigma - 4)\}^{-1}, \quad c_1 = (\varsigma - 3)c_2.$$

Using GN Theorem 3.3.18(ii), the first expectation on the right-hand-side of (S14) can be simplified as follows

$$\begin{aligned} E_S\{\text{tr}(\mathbf{e}_i \mathbf{e}_k^T \mathbf{S}^{-1}) \mathbf{S}^{-1}\} &= c_1 \text{tr}(\mathbf{e}_i \mathbf{e}_k^T \check{\mathbf{\Phi}}) \check{\mathbf{\Phi}} + c_2 (\check{\mathbf{\Phi}} \mathbf{e}_k \mathbf{e}_i^T \check{\mathbf{\Phi}} + \check{\mathbf{\Phi}} \mathbf{e}_i \mathbf{e}_k^T \check{\mathbf{\Phi}}) \\ &= c_1 \check{\phi}_{ik} \check{\mathbf{\Phi}} + c_2 (\check{\phi}^{(k)} \check{\phi}_i^T + \check{\phi}^{(i)} \check{\phi}_k^T). \end{aligned}$$

Using GN Theorem 3.3.18(i), the second expectation can be written

$$\begin{aligned} E_S(\mathbf{S}^{-1} \mathbf{e}_i \mathbf{e}_k^T \mathbf{S}^{-1}) &= c_1 \check{\mathbf{\Phi}} \mathbf{e}_i \mathbf{e}_k^T \check{\mathbf{\Phi}} + c_2 \left\{ \check{\mathbf{\Phi}} \mathbf{e}_k \mathbf{e}_i^T \check{\mathbf{\Phi}} + \text{tr}(\mathbf{e}_i \mathbf{e}_k^T \check{\mathbf{\Phi}}) \check{\mathbf{\Phi}} \right\} \\ &= c_1 \check{\phi}^{(i)} \check{\phi}_k^T + c_2 (\check{\phi}^{(k)} \check{\phi}_i^T + \check{\phi}_{ik} \check{\mathbf{\Phi}}), \end{aligned}$$

and, analogously, the third expectation becomes

$$E_S(\mathbf{S}^{-1} \mathbf{e}_k \mathbf{e}_i^T \mathbf{S}^{-1}) = c_1 \check{\phi}^{(k)} \check{\phi}_i^T + c_2 (\check{\phi}^{(i)} \check{\phi}_k^T + \check{\phi}_{ik} \check{\mathbf{\Phi}}).$$

Substituting these expressions back into (S14) gives

$$\begin{aligned} E(\mathbf{\Lambda} \mathbf{\Lambda}^T \mathbf{e}_i \mathbf{e}_k^T \mathbf{\Lambda} \mathbf{\Lambda}^T) &= \{\text{tr}(\mathbf{\Psi}^2)(c_1 + c_2) + \text{tr}(\mathbf{\Psi})^2 c_2\} (\check{\phi}_{ik} \check{\mathbf{\Phi}} + \check{\phi}^{(k)} \check{\phi}_i^T) \\ &\quad + \{2\text{tr}(\mathbf{\Psi}^2) c_2 + \text{tr}(\mathbf{\Psi})^2 c_1\} \check{\phi}^{(i)} \check{\phi}_k^T. \end{aligned}$$

Pre-multiplication by $\mathbf{e}_j^T$ and post-multiplication by $\mathbf{e}_\ell$ then yields

$$\begin{aligned} \mathbf{e}_j^T E(\mathbf{\Lambda} \mathbf{\Lambda}^T \mathbf{e}_i \mathbf{e}_k^T \mathbf{\Lambda} \mathbf{\Lambda}^T) \mathbf{e}_\ell &= \{\text{tr}(\mathbf{\Psi}^2)(c_1 + c_2) + \text{tr}(\mathbf{\Psi})^2 c_2\} (\check{\phi}_{il} \check{\phi}_{jk} + \check{\phi}_{ik} \check{\phi}_{jl}) + \{2\text{tr}(\mathbf{\Psi}^2) c_2 + \text{tr}(\mathbf{\Psi})^2 c_1\} \check{\phi}_{ij} \check{\phi}_{k\ell} \\ &= c_2 \{\text{tr}(\mathbf{\Psi}^2)(\varsigma - 2) + \text{tr}(\mathbf{\Psi})^2\} (\check{\phi}_{il} \check{\phi}_{jk} + \check{\phi}_{ik} \check{\phi}_{jl}) + c_2 \{2\text{tr}(\mathbf{\Psi}^2) + \text{tr}(\mathbf{\Psi})^2(\varsigma - 3)\} \check{\phi}_{ij} \check{\phi}_{k\ell}. \end{aligned}$$

Using (S13), the product of the second order moments in (S11) is given by

$$\mathbf{e}_j^T E(\mathbf{\Lambda} \mathbf{\Lambda}^T) \mathbf{e}_i \mathbf{e}_k^T E(\mathbf{\Lambda} \mathbf{\Lambda}^T) \mathbf{e}_\ell = \frac{\text{tr}(\mathbf{\Psi})^2}{(\varsigma - 2)^2} \check{\phi}_{ij} \check{\phi}_{k\ell}.$$

The covariance between δ_{ij} and $\delta_{k\ell}$ is therefore

$$\text{Cov}(\delta_{ij}, \delta_{k\ell}) = c_3 \{\text{tr}(\mathbf{\Psi})^2 + (\varsigma - 2)\text{tr}(\mathbf{\Psi}^2)\} \left\{ 2\check{\phi}_{ij} \check{\phi}_{k\ell} + (\varsigma - 2)(\check{\phi}_{ik} \check{\phi}_{jl} + \check{\phi}_{il} \check{\phi}_{jk}) \right\}$$

where

$$c_3 = c_2(\varsigma - 2)^{-1} = \{(\varsigma - 1)(\varsigma - 2)^2(\varsigma - 4)\}^{-1}.$$

In the special case where $i = k$ and $j = \ell$, the variance of δ_{ij} is given by

$$\text{Var}(\delta_{ij}) = c_3 \{\text{tr}(\mathbf{\Psi})^2 + (\varsigma - 2)\text{tr}(\mathbf{\Psi}^2)\} \left\{ \varsigma \check{\phi}_{ij}^2 + (\varsigma - 2)\check{\phi}_{ii} \check{\phi}_{jj} \right\}.$$

For parameter interpretation and for the derivation of the results in the next section, it is convenient to reparameterise the distribution for $\mathbf{\Lambda}$ and therefore $\mathbf{\Delta}$ in terms of $\mathbf{\Phi} = \check{\mathbf{\Phi}}/(\varsigma - 2)$. The mean then reduces to

$$E(\mathbf{\Delta}) = E_{\Lambda}(\mathbf{\Lambda}\mathbf{\Lambda}^T) = \text{tr}(\mathbf{\Psi})\mathbf{\Phi}, \quad (\text{S15})$$

and the variances and covariances become

$$\text{Cov}(\delta_{ij}, \delta_{k\ell}) = \frac{\{\text{tr}(\mathbf{\Psi})^2 + (\varsigma - 2)\text{tr}(\mathbf{\Psi}^2)\} \{2\phi_{ij}\phi_{k\ell} + (\varsigma - 2)(\phi_{ik}\phi_{j\ell} + \phi_{i\ell}\phi_{jk})\}}{(\varsigma - 1)(\varsigma - 4)},$$

and

$$\text{Var}(\delta_{ij}) = \frac{\{\text{tr}(\mathbf{\Psi})^2 + (\varsigma - 2)\text{tr}(\mathbf{\Psi}^2)\} \{\varsigma\phi_{ij}^2 + (\varsigma - 2)\phi_{ii}\phi_{jj}\}}{(\varsigma - 1)(\varsigma - 4)}. \quad (\text{S16})$$

We note that the mean in (S15) now has exactly the same form as that obtained under the matrix normal prior. More significantly, it will be straightforward to study the effects of varying the degree of freedom parameter ς on the variances and covariances while keeping the mean constant.

S2.5 Bounding the variance under the matrix- t prior

The Frobenius inner product of two matrices $\mathbf{A} \in \mathbb{R}^{n \times m}$ and $\mathbf{B} \in \mathbb{R}^{n \times m}$ can be defined as

$$\langle \mathbf{A}, \mathbf{B} \rangle_F = \text{tr}(\mathbf{A}^T \mathbf{B}). \quad (\text{S17})$$

We can therefore apply the Cauchy-Schwartz inequality with $\mathbf{A} = \mathbf{\Psi}$ and $\mathbf{B} = \mathbf{I}_k$ which yields $\text{tr}(\mathbf{\Psi})^2 \leq k\text{tr}(\mathbf{\Psi}^2)$. Similarly, the norm induced by (S17) – the Frobenius norm – of $\mathbf{A} \in \mathbb{R}^{n \times m}$ is defined by

$$\|\mathbf{A}\|_F = \sqrt{\text{tr}(\mathbf{A}^T \mathbf{A})}.$$

The Frobenius norm possesses the submultiplicative property [see, for example 7, Chapter 15]. This means that for $\mathbf{A} \in \mathbb{R}^{n \times m}$ and $\mathbf{B} \in \mathbb{R}^{m \times p}$ we have

$$\|\mathbf{AB}\|_F \leq \|\mathbf{A}\|_F \|\mathbf{B}\|_F.$$

We can therefore take $\mathbf{A} = \mathbf{\Psi}^{1/2}$ and $\mathbf{B} = \mathbf{\Psi}^{1/2}$ and square each side to give $\text{tr}(\mathbf{\Psi}^2) \leq \text{tr}(\mathbf{\Psi})^2$. Altogether, this implies

$$\frac{1}{k}\text{tr}(\mathbf{\Psi})^2 \leq \text{tr}(\mathbf{\Psi}^2) \leq \text{tr}(\mathbf{\Psi})^2,$$

and so we can bound the left-hand term in the numerator of the variance in (S16) between

$$\text{tr}(\mathbf{\Psi})^2(\varsigma + k - 2)/k \leq \text{tr}(\mathbf{\Psi})^2 + (\varsigma - 2)\text{tr}(\mathbf{\Psi}^2) \leq \text{tr}(\mathbf{\Psi})^2(\varsigma - 1). \quad (\text{S18})$$

Now, taking $m_{ij} = E(\delta_{ij}) = \text{tr}(\Psi)\phi_{ij}$ in (S16) gives

$$\text{Var}(\delta_{ij}) = \frac{\{\text{tr}(\Psi)^2 + (\varsigma - 2)\text{tr}(\Psi^2)\} \{\varsigma m_{ij}^2 + (\varsigma - 2)m_{ii}m_{jj}\}}{\text{tr}(\Psi)^2(\varsigma - 1)(\varsigma - 4)}$$

and so we can use (S18) to bound the variance between

$$\frac{(\varsigma + k - 2)\{\varsigma m_{ij}^2 + (\varsigma - 2)m_{ii}m_{jj}\}}{k(\varsigma - 1)(\varsigma - 4)} \leq \text{Var}(\delta_{ij}) \leq \frac{\varsigma m_{ij}^2 + (\varsigma - 2)m_{ii}m_{jj}}{\varsigma - 4}.$$

For fixed values of the means m_{ij} , the lower and upper bounds are rational functions in ς , each of which is defined and positive for all $\varsigma > 4$. The lower limit approaches the horizontal asymptote $(m_{ij}^2 + m_{ii}m_{jj})/k$ as $\varsigma \rightarrow \infty$ whilst the upper limit approaches the horizontal asymptote $(m_{ij}^2 + m_{ii}m_{jj})$. Moreover, $\varsigma = 4$ is a vertical asymptote. This means that the variance $\text{Var}(\delta_{ij})$ is bound between two functions that are finite as $\varsigma \rightarrow \infty$ whilst becoming infinite as ς tends to 4 from above. It follows that we can keep the mean constant whilst using the degree of freedom parameter ς to control the variance.

S3 Markov chain Monte Carlo sampling

S3.1 Gibbs samplers for static factor models

S3.1.1 Gibbs sampler with fixed level of truncation

Consider the (static) factor model. Suppose that the matrix normal prior from Section 3.2 has been chosen for Λ and that we have completed our prior specification by assuming the idiosyncratic variances are independent and such that $\sigma_j^{-2} \sim \text{Gam}(a_\sigma, b_\sigma)$ for $j = 1, \dots, p$ and that $\mu \sim N(\mathbf{m}_\mu, \mathbf{V}_\mu)$. Given a fixed level of truncation H , one iteration of the non-adaptive Gibbs sampler involves a sweep through the following steps:

1. For $i = 1, \dots, n$ sample

$$\eta_i | \dots \sim N_H \left\{ (\mathbf{I}_H + \Lambda^T \Sigma^{-1} \Lambda)^{-1} \Lambda^T \Sigma^{-1} (\mathbf{y}_i - \mu), (\mathbf{I}_H + \Lambda^T \Sigma^{-1} \Lambda)^{-1} \right\}.$$

2. Sample $\text{vec}(\Lambda) | \dots \sim N_{pH}(\mathbf{c}, \mathbf{D}^{-1})$ where

$$\mathbf{D} = (\mathbf{E}^T \mathbf{E} \otimes \Sigma^{-1}) + (\Psi \otimes \Phi)^{-1}$$

and

$$\mathbf{c} = \mathbf{D}^{-1}(\mathbf{E}^T \otimes \Sigma^{-1}) \text{vec}\{\mathbf{Y} - (\mathbf{1}_n \otimes \mu^T)\}$$

where $\mathbf{Y}$ is a $n \times p$ matrix whose i^{th} row is $\mathbf{y}_i^T$ and $\mathbf{E}$ is a $n \times H$ matrix whose i^{th} row is η_i^T .

3. For $j = 1, \dots, p$, sample

$$\sigma_j^{-2} | \dots \sim \text{Gam} \left\{ a_\sigma + \frac{n}{2}, b_\sigma + \frac{1}{2} \sum_{i=1}^n (y_{ij} - \mu_j - \boldsymbol{\lambda}_j^\top \boldsymbol{\eta}_i)^2 \right\}$$

where $\boldsymbol{\lambda}_j^\top$ is row j of $\mathbf{\Lambda}$.

4. Sample

$$\boldsymbol{\mu} | \dots \sim \text{N}_p \left\{ (\mathbf{P}_\mu + n\boldsymbol{\Sigma}^{-1})(\mathbf{P}_\mu \mathbf{m}_\mu + n\boldsymbol{\Sigma}^{-1} \bar{\mathbf{y}}_\eta), (\mathbf{P}_\mu + n\boldsymbol{\Sigma}^{-1})^{-1} \right\}$$

where $\bar{\mathbf{y}}_\eta$ is the sample mean vector of $\mathbf{Y} - \mathbf{E}\mathbf{\Lambda}^\top$.

5. Update any correlation parameters, $\boldsymbol{\vartheta}$, in $\boldsymbol{\Phi}$ from

$$\pi(\boldsymbol{\vartheta} | \dots) \propto \pi(\boldsymbol{\vartheta}) \pi(\mathbf{\Lambda} | \boldsymbol{\Phi}, \boldsymbol{\Psi})$$

using a Metropolis-Hastings step.

6. Sample

$$\varrho_1 | \dots \sim \text{Gam} \left(a_1 + \frac{nH}{2}, 1 + \frac{1}{2} \sum_{h=1}^H g_h^{(1)} \boldsymbol{\lambda}^{(h)\top} \boldsymbol{\Phi}^{-1} \boldsymbol{\lambda}^{(h)} \right)$$

and then, for $j = 2, \dots, H$, sample

$$\varrho_j | \dots \sim \text{Gam} \left(a_2 + \frac{n}{2}(H - j + 1), 1 + \frac{1}{2} \sum_{h=j}^H g_h^{(j)} \boldsymbol{\lambda}^{(h)\top} \boldsymbol{\Phi}^{-1} \boldsymbol{\lambda}^{(h)} \right)$$

where $\boldsymbol{\lambda}^{(h)}$ denotes column h of $\mathbf{\Lambda}$ and $g_h^{(j)} = \prod_{\ell=1, \ell \neq j}^h \varrho_\ell$ for $j = 1, \dots, H$.

If, instead, a matrix- t prior has been chosen for $\mathbf{\Lambda}$ with unknown degree of freedom parameter ς , we substitute $\mathbf{S}^{-1}$ in place of $\boldsymbol{\Phi}$ in steps 2 and 6, replace step 5 with

5. Update any correlation parameters, $\boldsymbol{\vartheta}$, in $\check{\boldsymbol{\Phi}}$ from

$$\pi(\boldsymbol{\vartheta} | \dots) \propto \pi(\boldsymbol{\vartheta}) \pi(\mathbf{S} | \varsigma, \check{\boldsymbol{\Phi}}),$$

where $\mathbf{S} | \varsigma, \check{\boldsymbol{\Phi}} \sim \text{W}_p(\varsigma + p - 1, \check{\boldsymbol{\Phi}}^{-1})$, using a Metropolis-Hastings step.

We then add the following joint Metropolis-Hastings update between steps 4 and 5

4.5. Update $(\mathbf{S}, \varsigma)$ jointly by simulating ς^* from a proposal distribution with density $q(\varsigma^* | \varsigma)$, for example a lognormal distribution with location parameter $\log \varsigma$, and then proposing $\mathbf{S}^*$ from its full conditional distribution given ς^* :

$$\mathbf{S}^* | \varsigma^*, \mathbf{\Lambda}, \check{\boldsymbol{\Phi}}, \boldsymbol{\Psi} \sim \text{W}_p \left\{ H + p + \varsigma^* - 1, (\mathbf{\Lambda} \boldsymbol{\Psi}^{-1} \mathbf{\Lambda}^\top + \check{\boldsymbol{\Phi}})^{-1} \right\}.$$

In the acceptance probability $\min\{1, A\}$ of the proposal, A simplifies to

$$\begin{aligned} A &= \frac{q_1(\zeta|\zeta^*)\pi(\mathbf{S}|\zeta, \mathbf{\Lambda}, \check{\mathbf{\Phi}}, \mathbf{\Psi})}{q_1(\zeta^*|\zeta)\pi(\mathbf{S}^*|\zeta^*, \mathbf{\Lambda}, \check{\mathbf{\Phi}}, \mathbf{\Psi})} \times \frac{\pi(\mathbf{\Lambda}|\mathbf{S}^*, \mathbf{\Psi})\pi(\mathbf{S}^*|\zeta^*, \check{\mathbf{\Phi}})\pi(\zeta^*)}{\pi(\mathbf{\Lambda}|\mathbf{S}, \mathbf{\Psi})\pi(\mathbf{S}|\zeta, \check{\mathbf{\Phi}})\pi(\zeta)} \\ &= \frac{q_1(\zeta|\zeta^*)\pi(\mathbf{S}|\zeta, \mathbf{\Lambda}, \check{\mathbf{\Phi}}, \mathbf{\Psi})}{q_1(\zeta^*|\zeta)\pi(\mathbf{S}^*|\zeta^*, \mathbf{\Lambda}, \check{\mathbf{\Phi}}, \mathbf{\Psi})} \times \frac{\pi(\mathbf{S}^*|\zeta^*, \mathbf{\Lambda}, \check{\mathbf{\Phi}}, \mathbf{\Psi})\pi(\mathbf{\Lambda}|\zeta^*, \check{\mathbf{\Phi}}, \mathbf{\Psi})\pi(\zeta^*)}{\pi(\mathbf{S}|\zeta, \mathbf{\Lambda}, \check{\mathbf{\Phi}}, \mathbf{\Psi})\pi(\mathbf{\Lambda}|\zeta, \check{\mathbf{\Phi}}, \mathbf{\Psi})\pi(\zeta)} \\ &= \frac{q_1(\zeta|\zeta^*)\pi(\mathbf{\Lambda}|\zeta^*, \check{\mathbf{\Phi}}, \mathbf{\Psi})\pi(\zeta^*)}{q_1(\zeta^*|\zeta)\pi(\mathbf{\Lambda}|\zeta, \check{\mathbf{\Phi}}, \mathbf{\Psi})\pi(\zeta)} \end{aligned}$$

where $\pi(\mathbf{\Lambda}|\zeta, \check{\mathbf{\Phi}}, \mathbf{\Psi})$ is the density of a matrix- t random variable, $\mathbf{\Lambda}|\zeta, \check{\mathbf{\Phi}}, \mathbf{\Psi} \sim \mathbf{t}_{p,k}(\zeta, \mathbf{0}, \check{\mathbf{\Phi}}, \mathbf{\Psi})$, given by

$$p(\mathbf{\Lambda}) = \frac{\Gamma_k\{(\zeta + k + p - 1)/2\}}{\pi^{pk/2}\Gamma_k\{(\zeta + k - 1)/2\}|\check{\mathbf{\Psi}}|^{p/2}|\check{\mathbf{\Phi}}|^{k/2}} \det\left(\mathbf{I}_p + \check{\mathbf{\Phi}}^{-1}\mathbf{\Lambda}\check{\mathbf{\Psi}}^{-1}\mathbf{\Lambda}^\top\right)^{-(\zeta+k+p-1)/2}.$$

Here $\Gamma_k(\cdot)$ denotes the multivariate gamma function, which can be expressed as $\Gamma_k(x) = \pi^{k(k-1)/4} \prod_{i=1}^k \Gamma\{x - \frac{1}{2}(i-1)\}$ if $\text{Re}(x) > (k-1)/2$. We note that A does not depend on $\mathbf{S}^*$ and so $\mathbf{S}$ need only be sampled if the proposed value of ζ is accepted.

S3.1.2 Adaptive Gibbs sampler

Denote by $x^{[i]}$ the value of the unknown x on iteration i of the Gibbs sampler and suppose that adaptation is allowed after $\bar{i}$ iterations. Recall that we define the number of effective factors on iteration i as $k^{*[i]} = H - m^{[i]}$ where $m^{[i]}$ is the number of columns of loadings that do not contribute appreciably to $\mathbf{\Omega}^{[i]}$ according to a truncation criterion.

Iteration i of the adaptive Gibbs sampler begins with a sweep through all the steps of the non-adaptive algorithm above. Then if $i \geq \bar{i}$, sample $u \sim \text{U}(0, 1)$ and if $u < p(i)$ where $p(i) = \exp(\alpha_0 + \alpha_1 i)$, an adaptation step is carried out:

7. Compare $k^{*[i]}$ to H :

- (a) If $k^{*[i]} < H$, H is reduced to $k^{*[i]}$ and any inactive factors are deleted along with the corresponding columns of $\mathbf{\Lambda}$ and components of $\mathbf{\varrho}$ and $\mathbf{\Psi}$ from the multiplicative gamma process (MGP) prior.
- (b) If $k^{*[i]} = H$ and $H < \lceil \varphi(p) \rceil - 1$, an extra factor, component of $\mathbf{\varrho}$ and column of factor loadings are sampled from their priors. That is, we sample $\eta_{i,H+1} \sim \text{N}(0, 1)$ ($i = 1, \dots, n$) and $\varrho_{H+1} \sim \text{Gam}(a_1, 1)$ if $H = 0$ or $\varrho_{H+1} \sim \text{Gam}(a_2, 1)$ otherwise. We then compute $\psi_{H+1} = 1/\varrho_{H+1}$ if $H = 0$ or $\psi_{H+1} = \psi_H/\varrho_{H+1}$ otherwise, and sample $\lambda_{(1),H+1} \sim \text{N}_p(\mathbf{0}, \psi_{H+1}\check{\mathbf{\Phi}})$. Finally, we increase H by 1.

S3.2 Gibbs samplers for dynamic factor models

S3.2.1 Gibbs sampler with fixed level of truncation

Consider the stationary dynamic factor model presented through equations (10) and (11) of the main manuscript and recall that $\mathbf{G}_i = \text{Cov}(\boldsymbol{\eta}_t, \boldsymbol{\eta}_{t+i})$ is the i^{th}

autocovariance of $\boldsymbol{\eta}_t$. The joint distribution of the factors can be expressed as

$$\begin{aligned} \pi(\boldsymbol{\eta}_{(1-m):n} | \boldsymbol{\Gamma}_1, \dots, \boldsymbol{\Gamma}_m, \boldsymbol{\Pi}) &= \pi(\boldsymbol{\eta}_{(1-m):0} | \boldsymbol{\Gamma}_1, \dots, \boldsymbol{\Gamma}_m, \boldsymbol{\Pi}) \\ &\times \prod_{t=m+1}^n \pi(\boldsymbol{\eta}_t | \boldsymbol{\eta}_{(t-p):(t-1)}, \boldsymbol{\Gamma}_1, \dots, \boldsymbol{\Gamma}_m, \boldsymbol{\Pi}) \end{aligned} \quad (\text{S19})$$

in which $\boldsymbol{\eta}_t | \boldsymbol{\eta}_{(t-m):(t-1)}, \boldsymbol{\Gamma}_1, \dots, \boldsymbol{\Gamma}_m, \boldsymbol{\Pi} \sim \text{N}_k(\sum_{i=1}^m \boldsymbol{\Gamma}_i \boldsymbol{\eta}_{t-i}, \boldsymbol{\Pi})$ and the initial distribution is $(\boldsymbol{\eta}_{1-m}^\top, \dots, \boldsymbol{\eta}_0^\top)^\top | \boldsymbol{\Gamma}_1, \dots, \boldsymbol{\Gamma}_m, \boldsymbol{\Pi} \sim \text{N}_{km}(\mathbf{0}, \boldsymbol{G})$ where $\boldsymbol{G}$ is a positive definite block Toeplitz matrix with $\boldsymbol{G}_{j-i}$ as the block in rows $\{k(i-1)+1\}$ to ki and columns $\{k(j-1)+1\}$ to kj ($i, j = 1, \dots, m$) and $\boldsymbol{G}_{-\ell} = \boldsymbol{G}_\ell^\top$ ($\ell = 1, \dots, m-1$). For the purposes of evaluating (S19), the stationary variance $\boldsymbol{G}_0$ is fixed at $\boldsymbol{I}_k$ and the autocovariances $\boldsymbol{G}_1, \dots, \boldsymbol{G}_{m-1}$, along with the innovation variance $\boldsymbol{\Pi}$, are available as by-products of the reverse mapping from the unconstrained parameters $(\boldsymbol{A}_1, \dots, \boldsymbol{A}_m)$ to $(\boldsymbol{\Gamma}_1, \dots, \boldsymbol{\Gamma}_m)$; see the Appendix in Heaps [8] for full details. The original parameters $(\boldsymbol{\Gamma}_1, \dots, \boldsymbol{\Gamma}_m, \boldsymbol{\Pi})$ are therefore a complicated function of the unconstrained parameters $\boldsymbol{A}_1, \dots, \boldsymbol{A}_m$ and so the full conditional density of any particular $\boldsymbol{A}_j$ can be written as

$$\pi(\boldsymbol{A}_j | \boldsymbol{A}_{-j}, \boldsymbol{\eta}_{(1-m):n}) \propto \pi(\boldsymbol{A}_j) \pi(\boldsymbol{\eta}_{(1-m):n} | \boldsymbol{A}_1, \dots, \boldsymbol{A}_m), \quad (\text{S20})$$

in which $\boldsymbol{A}_{-j}$ denotes the collection $(\boldsymbol{A}_1, \dots, \boldsymbol{A}_m)$ with $\boldsymbol{A}_j$ removed and $\boldsymbol{A}_j \sim \text{N}_{k,k}(\mathbf{0}, \boldsymbol{I}_k, \boldsymbol{I}_k)$.

In order to extend the non-adaptive algorithm in the previous section to the dynamic case, we replace step 1 with

1. Sample $\boldsymbol{\eta}_{1-m}, \dots, \boldsymbol{\eta}_n$ from their full conditional distribution as a single block using a forward-filtering backward-sampling algorithm [e.g. 9, 10].

We also add the following step between steps 2 and 3

- 2.5. For $j = 1, \dots, m$ update $\text{vec}(\boldsymbol{A}_j)$ from its full conditional density (S20) using a Metropolis-adjusted Langevin (MALA) proposal

$$\text{vec}(\boldsymbol{A}_j^*) | \text{vec}(\boldsymbol{A}_j) \sim \text{N}_{H^2} \left\{ \text{vec}(\boldsymbol{A}_j) + \frac{s^2}{2} \nabla \log \pi(\boldsymbol{A}_j | \boldsymbol{A}_{-j}, \boldsymbol{\eta}_{(1-m):n}), s^2 \boldsymbol{I}_{H^2} \right\}$$

in which $\nabla \log \pi(\boldsymbol{A}_j | \boldsymbol{A}_{-j}, \boldsymbol{\eta}_{(1-m):n}) = \partial \log \pi(\boldsymbol{A}_j | \boldsymbol{A}_{-j}, \boldsymbol{\eta}_{(1-m):n}) / \partial \{\text{vec}(\boldsymbol{A}_j)\}^\top$ is the gradient of the logarithm of the full conditional density for $\boldsymbol{A}_j$ and s^2 is a tuning variance.

If the acceptance rate is too small for reasonably sized s , we can divide each $\boldsymbol{A}_j$ into blocks of length b and use the approach above to update them one-at-a-time.

When the order of the vector autoregression in the state equation is $m = 1$, $\nabla \log \pi(\boldsymbol{A}_j | \boldsymbol{A}_{-j}, \boldsymbol{\eta}_{(1-m):n})$ can be calculated analytically; full details are given at the end of Section S3.2. When $m > 1$, the gradients can be evaluated with automatic differentiation. Alternatively, the MALA steps can be replaced with Gaussian random walks.

S3.2.2 Adaptive Gibbs sampler

As in the static case, the adaptive Gibbs sampler for a dynamic model begins with a sweep through all the steps of the non-adaptive scheme. In the adaptation step, an extra complication arises when the truncation point H is increased by 1. Ideally, we would add a row and column to the parameters of the state equation, $\mathbf{\Gamma}_1, \dots, \mathbf{\Gamma}_m$, by sampling from their prior, conditional on the submatrices that already occupy the first H rows and columns. However, in order to enforce stationarity, we work with a reparameterisation of the vector autoregressive model for the factors. Priors are placed on the unconstrained parameters $\mathbf{A}_1, \dots, \mathbf{A}_m$ on which the parameters in the original model, $\mathbf{\Gamma}_1, \dots, \mathbf{\Gamma}_m, \mathbf{\Pi}$, depend. Due to the recursive nature of the mapping between the original parameters and the intermediate partial autocorrelation parameterisation $\mathbf{P}_1, \dots, \mathbf{P}_m$, the Jacobian of the transformation is not available in closed form and so neither is the prior for the original parameters over the stationary region. We cannot, therefore, simply sample extra rows and columns of the $\mathbf{\Gamma}_i$ from their prior.

As an alternative, we propose basing the adaptation on the parsimonious assumption that the new factor does not interact with the existing factors at lags $1, \dots, m$ and interacts minimally with itself. This requires filling the first H elements in row and column $H + 1$ of the adapted matrices $\mathbf{\Gamma}_1, \dots, \mathbf{\Gamma}_m$ with zeros. Equivalently, we can fill the first H elements in row and column $H + 1$ of the adapted partial autocorrelation matrices $\mathbf{P}_1, \dots, \mathbf{P}_m$ with zeros. The partial autocorrelation matrices lie in the space of square matrices whose singular values are all less than one. We can therefore guarantee a new parameter set that lies in the stationary region whilst adhering to our adaptation assumptions by simulating the $(H + 1, H + 1)$ element in the adapted $\mathbf{P}_i$ by sampling uniformly on $(0, r_{i,\min})$ where $r_{i,\min}$ is the smallest singular value of $\mathbf{P}_{i,1:H,1:H}$. We can then compute the corresponding value for $\mathbf{\Pi}$ using step 2 of the reverse mapping in the Appendix of Heaps [8].

Altogether, the adaptation step of the Gibbs sampler for a dynamic model replaces step 7 in the static case with the following:

7. Compare $k^{*[i]}$ to H :

- (a) If $k^{*[i]} < H$, H is reduced to $k^{*[i]}$ and any inactive factors are deleted along with the corresponding columns of $\mathbf{\Lambda}$, columns and rows of $(\mathbf{\Gamma}_1, \dots, \mathbf{\Gamma}_m, \mathbf{\Pi})$, and components of $\mathbf{\varrho}$ and $\mathbf{\Psi}$ from the MGP prior. We then recompute $(\mathbf{A}_1, \dots, \mathbf{A}_m)$.
- (b) If $k^{*[i]} = H$ and $H < \lceil \varphi(p) \rceil - 1$, an extra component of $\mathbf{\varrho}$ and column of factor loadings are sampled from their priors. That is, we sample $\varrho_{H+1} \sim \text{Gam}(a_1, 1)$ if $H = 0$ or $\varrho_{H+1} \sim \text{Gam}(a_2, 1)$ otherwise. We then compute $\psi_{H+1} = 1/\varrho_{H+1}$ if $H = 0$ or $\psi_{H+1} = \psi_H/\varrho_{H+1}$ otherwise, and sample $\boldsymbol{\lambda}_{(1),H+1} \sim \text{N}_p(\mathbf{0}, \psi_{H+1}\mathbf{\Phi})$. For $i = 1, \dots, m$, we set $p_{i,H+1,j} = p_{i,j,H+1} = 0$ for $j = 1, \dots, H$ and sample $p_{i,H+1,H+1} \sim \text{U}(0, r_{i,\min})$. We then recompute $(\mathbf{A}_1, \dots, \mathbf{A}_m)$ and $\mathbf{\Pi}$. We additionally compute $(\mathbf{\Gamma}_1, \dots, \mathbf{\Gamma}_m)$ and $\mathbf{G}$ so that we can sample an extra factor $(\eta_{1-m,H+1}, \dots, \eta_{n,H+1})^\top$ from the implied univariate autoregressive model of order m . That is, we sample $(\eta_{1-m,H+1}, \dots, \eta_{0,H+1})^\top \sim \text{N}_m(\mathbf{0}, \mathbf{G}^*)$ and then, for $t = 1, \dots, n$, $\eta_{t,H+1} \sim \text{N}(\sum_{i=1}^m \gamma_{i,H+1,H+1} \eta_{t-i,H+1}, \pi_{H+1,H+1})$ where $\mathbf{G}^*$ comprises the submatrix of $\mathbf{G}$ made up of rows and columns $(H+1, 2H+2, \dots, mH+m)$. Finally, we increase H by 1.

S3.2.3 Gradient of the logarithm of the full conditional density when $m = 1$

In the special case when $m = 1$, there are closed form expressions for the lag-1 autoregressive coefficient matrix $\mathbf{\Gamma}_1$ and the innovation variance $\mathbf{\Pi}$ in terms of the transformed partial autocorrelation matrix $\mathbf{A}_1$. Dropping the 1-subscript for brevity, we can write $\mathbf{\Gamma} = (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{-1/2}\mathbf{A}$ and $\mathbf{\Pi} = (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{-1}$. This allows the joint density of the factors (S19), and hence the logarithm of the full conditional density for $\mathbf{A}$, to be written as a closed form function of $\mathbf{A}$. Up to an additive constant of proportionality, we have

$$\begin{aligned} \log \pi(\mathbf{A}|\boldsymbol{\eta}_{0:n}) &= \frac{n}{2} \log |\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top| - \frac{1}{2} \text{tr}(\mathbf{A}\mathbf{A}^\top) \\ &\quad - \frac{1}{2} \sum_{t=1}^n \left\{ \boldsymbol{\eta}_t^\top \mathbf{A}\mathbf{A}^\top \boldsymbol{\eta}_t - 2\boldsymbol{\eta}_t^\top (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \mathbf{A}\boldsymbol{\eta}_{t-1} + \boldsymbol{\eta}_{t-1}^\top \mathbf{A}\mathbf{A}^\top \boldsymbol{\eta}_{t-1} \right\}. \end{aligned} \quad (\text{S21})$$

whose gradient can be derived analytically.

Theorem S2. *The gradient of the logarithm of the full conditional density in (S21) is given by*

$$\begin{aligned} \frac{\partial \log \pi(\mathbf{A}|\boldsymbol{\eta}_{0:n})}{\partial \{\text{vec}(\mathbf{A})\}^\top} &= \frac{n}{2} \text{vec} \left\{ (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{-1} \right\}^\top (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)}) (\mathbf{A} \otimes \mathbf{I}_k) - \text{vec}(\mathbf{A})^\top \\ &\quad - \frac{1}{2} \sum_{t=1}^n \left((\boldsymbol{\eta}_t^\top \otimes \boldsymbol{\eta}_t^\top) (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)}) (\mathbf{A} \otimes \mathbf{I}_k) + (\boldsymbol{\eta}_{t-1}^\top \otimes \boldsymbol{\eta}_{t-1}^\top) (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)}) (\mathbf{I}_k \otimes \mathbf{A}^\top) \right. \\ &\quad \left. - 2(\boldsymbol{\eta}_{t-1}^\top \otimes \boldsymbol{\eta}_t^\top) \left[\mathbf{I}_k \otimes (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \right. \right. \\ &\quad \left. \left. + (\mathbf{A} \otimes \mathbf{I}_k)^\top \{ (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \oplus (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \}^{-1} (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)}) (\mathbf{A} \otimes \mathbf{I}_k) \right] \right). \end{aligned}$$

Proof. In order to compute the gradient of (S21) we make extensive use of the results from Chapter 17 of Seber [4], as detailed below. From Result 17.30(g),

$$\frac{\partial}{\partial \text{vec}(\mathbf{A})^\top} \text{vec}(\mathbf{A}\mathbf{A}^\top) = (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)}) (\mathbf{A} \otimes \mathbf{I}_k)$$

and so, using Result 17.33(a) yields

$$\begin{aligned} \frac{\partial}{\partial \text{vec}(\mathbf{A})^\top} (\boldsymbol{\eta}_t^\top \mathbf{A}\mathbf{A}^\top \boldsymbol{\eta}_t) &= (\boldsymbol{\eta}_t^\top \otimes \boldsymbol{\eta}_t^\top) \frac{\partial}{\partial \text{vec}(\mathbf{A})^\top} \text{vec}(\mathbf{A}\mathbf{A}^\top) \\ &= (\boldsymbol{\eta}_t^\top \otimes \boldsymbol{\eta}_t^\top) (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)}) (\mathbf{A} \otimes \mathbf{I}_k) \end{aligned} \quad (\text{S22})$$

in which $\mathbf{I}_{(k,k)}$ denotes a vec-permutation (or commutation) matrix, that is, the $k^2 \times k^2$ permutation matrix such that $\text{vec}(\mathbf{A}) = \mathbf{I}_{(k,k)} \text{vec}(\mathbf{A}^\top)$.

Similarly, using Result 17.30(f)

$$\frac{\partial}{\partial \text{vec}(\mathbf{A})^T} \text{vec}(\mathbf{A}^T \mathbf{A}) = (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)})(\mathbf{I}_k \otimes \mathbf{A}^T)$$

and so, using Result 17.33(a) again yields

$$\begin{aligned} \frac{\partial}{\partial \text{vec}(\mathbf{A})^T} (\eta_{t-1}^T \mathbf{A}^T \mathbf{A} \eta_{t-1}) &= (\eta_{t-1}^T \otimes \eta_{t-1}^T) \frac{\partial}{\partial \text{vec}(\mathbf{A})^T} \text{vec}(\mathbf{A}^T \mathbf{A}) \\ &= (\eta_{t-1}^T \otimes \eta_{t-1}^T)(\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)})(\mathbf{I}_k \otimes \mathbf{A}^T). \end{aligned} \quad (\text{S23})$$

From Results 17.30(h) and 17.30(a), we have

$$\begin{aligned} &\frac{\partial}{\partial \text{vec}(\mathbf{A})^T} \text{vec} \left\{ (\mathbf{I}_k + \mathbf{A} \mathbf{A}^T)^{1/2} \mathbf{A} \right\} \\ &= (\mathbf{A} \otimes \mathbf{I}_k)^T \frac{\partial}{\partial \text{vec}(\mathbf{A})^T} \text{vec} \left\{ (\mathbf{I}_k + \mathbf{A} \mathbf{A}^T)^{1/2} \right\} + \left\{ \mathbf{I}_k \otimes (\mathbf{I}_k + \mathbf{A} \mathbf{A}^T)^{1/2} \right\} \frac{\partial \text{vec} \mathbf{A}}{\partial \text{vec}(\mathbf{A})^T} \\ &= (\mathbf{A} \otimes \mathbf{I}_k)^T \frac{\partial}{\partial \text{vec}(\mathbf{A})^T} \text{vec} \left\{ (\mathbf{I}_k + \mathbf{A} \mathbf{A}^T)^{1/2} \right\} + \mathbf{I}_k \otimes (\mathbf{I}_k + \mathbf{A} \mathbf{A}^T)^{1/2}. \end{aligned} \quad (\text{S24})$$

Now, using Result 17.31, with $\mathbf{F}(\mathbf{A}) = \mathbf{Z}\{\mathbf{Y}(\mathbf{A})\}$ where $\mathbf{Z}(\mathbf{V}) = \mathbf{V}^{1/2}$ and $\mathbf{Y}(\mathbf{A}) = \mathbf{I}_k + \mathbf{A} \mathbf{A}^T$, we have

$$\frac{\partial \text{vec}(\mathbf{F})}{\partial \text{vec}(\mathbf{A})^T} = \frac{\partial \text{vec}\{\mathbf{Z}(\mathbf{V})\}}{\partial \{\text{vec}(\mathbf{V})\}^T} \bigg|_{\mathbf{V}=\mathbf{Y}(\mathbf{A})} \frac{\partial \text{vec}\{\mathbf{Y}(\mathbf{A})\}}{\partial \text{vec}(\mathbf{A})^T}. \quad (\text{S25})$$

Using Result 17.30(g) again we have

$$\frac{\partial \text{vec}\{\mathbf{Y}(\mathbf{A})\}}{\partial \text{vec}(\mathbf{A})^T} = (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)})(\mathbf{A} \otimes \mathbf{I}_k). \quad (\text{S26})$$

In order to compute $\partial \text{vec}\{\mathbf{Z}(\mathbf{V})\} / \partial \{\text{vec}(\mathbf{V})\}^T = \partial \text{vec}(\mathbf{V}^{1/2}) / \partial \{\text{vec}(\mathbf{V})\}^T$ we first note that

$$\mathbf{V}^{1/2} \mathbf{V}^{1/2} = \mathbf{V}.$$

Taking the differential of each side and noting that $\mathbf{V}^{1/2}$ is a symmetric matrix yields

$$\mathbf{V}^{1/2} d\mathbf{V}^{1/2} + (d\mathbf{V}^{1/2}) \mathbf{V}^{1/2} = d\mathbf{V}.$$

This can be regarded as a Sylvester equation in $d\mathbf{V}^{1/2}$ which can be rewritten as

$$(\mathbf{I}_k \otimes \mathbf{V}^{1/2} + \mathbf{V}^{1/2} \otimes \mathbf{I}_k) \text{vec}(d\mathbf{V}^{1/2}) = (\mathbf{V}^{1/2} \oplus \mathbf{V}^{1/2}) \text{vec}(d\mathbf{V}^{1/2}) = \text{vec}(d\mathbf{V})$$

in which $\oplus$ is the Kronecker sum operator. Using Result 17.56(e) we can write this as

$$(\mathbf{V}^{1/2} \oplus \mathbf{V}^{1/2}) d\text{vec}(\mathbf{V}^{1/2}) = d\text{vec}(\mathbf{V}).$$

Since $\mathbf{V}^{1/2}$ is positive definite, $(\mathbf{V}^{1/2} \oplus \mathbf{V}^{1/2})$ will also be positive definite and so

$$d\text{vec}(\mathbf{V}^{1/2}) = (\mathbf{V}^{1/2} \oplus \mathbf{V}^{1/2})^{-1} d\text{vec}(\mathbf{V}).$$

We can then use Result 17.58 to obtain

$$\frac{\partial \text{vec}(\mathbf{V}^{1/2})}{\partial \text{vec}(\mathbf{V})^\top} = (\mathbf{V}^{1/2} \oplus \mathbf{V}^{1/2})^{-1}. \quad (\text{S27})$$

Taking (S26) and (S27) in (S25) gives

$$\frac{\partial}{\partial \text{vec}(\mathbf{A})^\top} \text{vec} \left\{ (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \right\} = \{ (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \oplus (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \}^{-1} (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)}) (\mathbf{A} \otimes \mathbf{I}_k)$$

which can be substituted into (S24) to obtain

$$\begin{aligned} \frac{\partial}{\partial \text{vec}(\mathbf{A})^\top} \text{vec} \left\{ (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \mathbf{A} \right\} &= (\mathbf{A} \otimes \mathbf{I}_k)^\top \{ (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \oplus (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \}^{-1} \\ &\quad \times (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)}) (\mathbf{A} \otimes \mathbf{I}_k) + \mathbf{I}_k \otimes (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2}. \end{aligned}$$

Finally, from Result 17.33(a) we have

$$\begin{aligned} \frac{\partial}{\partial \text{vec}(\mathbf{A})^\top} \left\{ \eta_t^\top (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \mathbf{A} \eta_{t-1} \right\} \\ &= (\eta_{t-1}^\top \otimes \eta_t^\top) \frac{\partial}{\partial \text{vec}(\mathbf{A})^\top} \text{vec} \left\{ (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \mathbf{A} \right\} \\ &= (\eta_{t-1}^\top \otimes \eta_t^\top) \left[(\mathbf{A} \otimes \mathbf{I}_k)^\top \{ (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \oplus (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \}^{-1} (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)}) (\mathbf{A} \otimes \mathbf{I}_k) \right. \\ &\quad \left. + \mathbf{I}_k \otimes (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{1/2} \right]. \end{aligned} \quad (\text{S28})$$

Denoting by $\text{adj}(\mathbf{X})$ the adjoint of a square matrix $\mathbf{X}$, using Result 17.26(c) with Result 17.30(g) gives

$$\begin{aligned} \frac{\partial}{\partial \text{vec}(\mathbf{A})^\top} |\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top| &= \text{vec} \left[\{ \text{adj}(\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top) \}^\top \right]^\top \frac{\partial}{\partial \text{vec}(\mathbf{A})^\top} \text{vec}(\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top) \\ &= \text{vec} \left\{ |\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top| (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{-1} \right\}^\top (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)}) (\mathbf{A} \otimes \mathbf{I}_k) \end{aligned}$$

and then applying the Chain Rule (Result 17.21) gives

$$\begin{aligned} \frac{\partial}{\partial \text{vec}(\mathbf{A})^\top} \log |\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top| \\ &= \frac{1}{|\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top|} \text{vec} \left\{ |\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top| (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{-1} \right\}^\top (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)}) (\mathbf{A} \otimes \mathbf{I}_k) \\ &= \text{vec} \left\{ (\mathbf{I}_k + \mathbf{A}\mathbf{A}^\top)^{-1} \right\}^\top (\mathbf{I}_{k^2} + \mathbf{I}_{(k,k)}) (\mathbf{A} \otimes \mathbf{I}_k). \end{aligned} \quad (\text{S29})$$

Finally, for the contribution from the prior, we can compute

$$\frac{\partial}{\partial \text{vec}(\mathbf{A})} \left\{ -\frac{1}{2} \text{tr}(\mathbf{A}\mathbf{A}^\top) \right\} = -\text{vec}(\mathbf{A})$$

which follows from Result 17.23. Therefore

$$\frac{\partial}{\partial \text{vec}(\mathbf{A})^\top} \left\{ -\frac{1}{2} \text{tr}(\mathbf{A}\mathbf{A}^\top) \right\} = -\text{vec}(\mathbf{A})^\top. \quad (\text{S30})$$

Combining (S22), (S23), (S28), (S29) and (S30) gives the required gradient. $\square$

S3.3 Post-processing the posterior output

Although the MCMC samplers described in this work generate draws from the posterior of the unknowns in the unidentified and unconstrained parameterisation of the factor model, an identified parameterisation is incredibly valuable for parameter interpretation and the assessment of MCMC diagnostics. Fortunately, samples of the unknowns in the unidentified and unconstrained parameterisation can be post-processed to obtain samples under a parameterisation in which the factor loadings matrix is identified by the PLT constraint. Denote by $\tilde{\boldsymbol{\eta}}_i$ the factors in an identified model with corresponding factor loadings matrix $\tilde{\mathbf{\Lambda}}$ which satisfies the PLT condition. In the dynamic model, additionally denote by $\tilde{\mathbf{\Gamma}}_1, \dots, \tilde{\mathbf{\Gamma}}_m$ and $\tilde{\mathbf{\Pi}}$ the autoregressive coefficients and innovation variance in the corresponding state equation, and by $\tilde{\mathbf{A}}_1, \dots, \tilde{\mathbf{A}}_m$ the transformed partial autocorrelation matrices. Now, for each sample from the posterior, a set of identifiable parameters can be found by calculating the LQ decomposition of $\mathbf{\Lambda}$, $\mathbf{\Lambda} = \tilde{\mathbf{\Lambda}}\mathbf{Q}$, and then transforming $\tilde{\boldsymbol{\eta}}_i = \mathbf{Q}\boldsymbol{\eta}_i$ for $i = 1, \dots, n$ and, in the dynamic case, $\tilde{\mathbf{\Gamma}}_i = \mathbf{Q}\mathbf{\Gamma}_i\mathbf{Q}^\top$, $i = 1, \dots, m$, and $\tilde{\mathbf{\Pi}} = \mathbf{Q}\mathbf{\Pi}\mathbf{Q}^\top$. Equivalently, in terms of the unconstrained reparameterisation of the stationary state equation, $\tilde{\mathbf{A}}_i = \mathbf{Q}\mathbf{A}_i\mathbf{Q}^\top$ for $i = 1, \dots, m$.

In order to illustrate the importance of basing MCMC diagnostics on the identified parameters, Figure S1 shows some graphical diagnostics from the analysis of the Finnish bird data in Section 6.2 of the main manuscript. In these plots the four different colours correspond to the output under four chains: two adaptive Gibbs chains and two Hamiltonian Monte Carlo chains implemented using Stan. The first two columns show diagnostics for two selected parameters from the unconstrained factor loadings matrix, namely $\lambda_{8,1}$ and $\lambda_{27,1}$. The second two columns show diagnostics for parameters in the corresponding positions of the identified factor loadings matrix, $\tilde{\lambda}_{8,1}$ and $\tilde{\lambda}_{27,1}$. The lack of identifiability of the unconstrained matrix is very clear. The two marginal posterior densities are multimodal with two major modes that are symmetric about zero. These arise because both the likelihood and prior are invariant to a flip of the sign of each column of the unconstrained factor loadings matrix. Some of the chains find and remain in one of the two modes, others occasionally jump between modes. In terms of the factor model, these modes are equivalent; indeed, we see no evidence of any lack of convergence or problems with mixing in the diagnostics for the identified parameters, and this was true for all other unknowns in $\tilde{\mathbf{\Lambda}}$. However, it

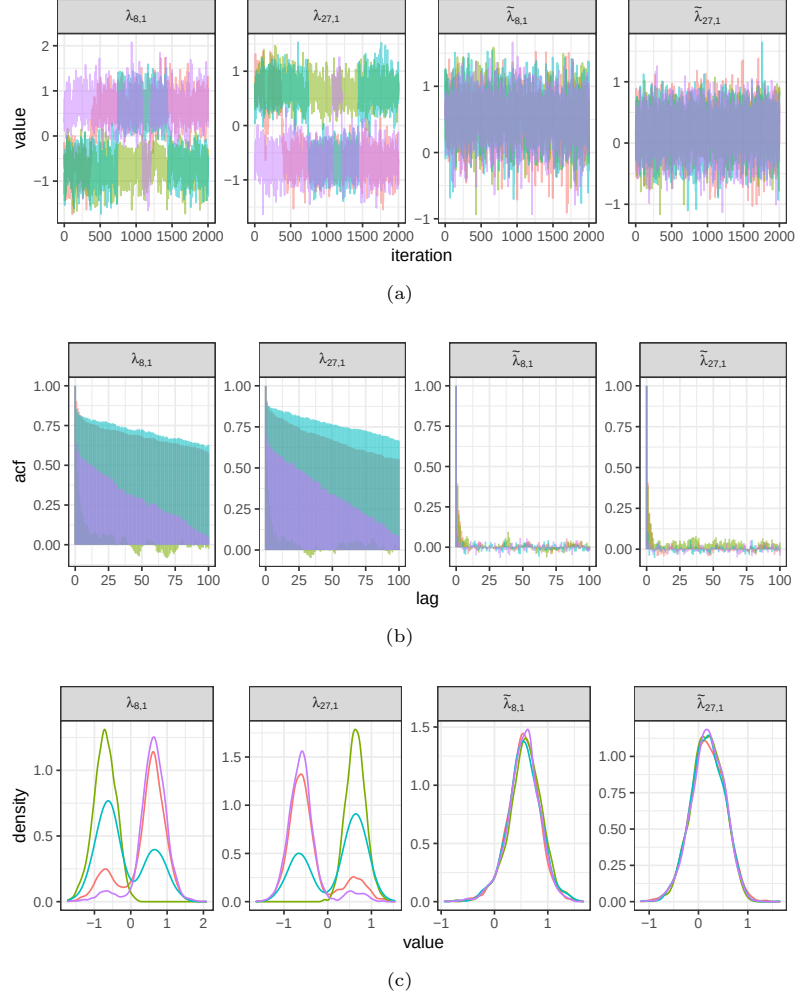


Fig. S1: Graphical diagnostics for two parameters from the unconstrained factor loadings matrix ($\lambda_{8,1}$ and $\lambda_{27,1}$) and two parameters from the factor loadings matrix identified by the PLT constraint ($\tilde{\lambda}_{8,1}$ and $\tilde{\lambda}_{27,1}$): (a) trace plots, (b) autocorrelation plots and (c) marginal density plots. Different colours depict the output from two Gibbs chains (—, —) and two Hamiltonian Monte Carlo chains run using Stan (—, —).

demonstrates the importance of basing diagnostic checks on samples of the identifiable parameters $\tilde{\Lambda}$ rather than the non-identifiable parameters Λ ; using samples of Λ , one cannot distinguish confidently between artefacts of non-identifiability and failed convergence of the sampler over the identifiable parameter space.

Table S1: Summaries of the simulation experiment when $\beta = 0.95$.

Prior	(p, k)	$E\{d_G(\boldsymbol{\Omega}, \boldsymbol{\Omega}_0) \mathbf{y}\}$		$E(k^* \mathbf{y})$	
		Median	IQR	Median	IQR
SN (expcov.)	(24, 6)	11.19	0.41	6.57	0.94
SN (fixed)	(24, 6)	10.70	0.72	6.29	0.65
SN (exch.)	(24, 6)	10.86	0.60	6.29	0.49
MGP	(24, 6)	11.29	0.60	5.71	1.70
CUSP	(24, 6)	11.10	0.57	3.02	1.09
SN (expcov.)	(48, 9)	14.88	0.97	7.92	1.36
SN (fixed)	(48, 9)	14.92	0.93	7.85	1.87
SN (exch.)	(48, 9)	15.04	0.97	6.15	1.53
MGP	(48, 9)	15.24	1.25	6.00	2.17
CUSP	(48, 9)	15.14	1.11	3.00	0.03
SN (expcov.)	(72, 9)	17.10	0.83	7.20	1.36
SN (fixed)	(72, 9)	16.92	0.89	7.41	1.45
SN (exch.)	(72, 9)	17.24	0.61	5.73	1.10
MGP	(72, 9)	17.60	0.80	6.20	1.28
CUSP	(72, 9)	17.52	0.88	3.00	0.00

Table S2: Summaries of the simulation experiment when $\beta = 0.9$.

Prior	(p, k)	$E\{d_G(\boldsymbol{\Omega}, \boldsymbol{\Omega}_0) \mathbf{y}\}$		$E(k^* \mathbf{y})$	
		Median	IQR	Median	IQR
SN (expcov.)	(24, 6)	9.33	0.34	7.03	0.58
SN (fixed)	(24, 6)	8.96	0.47	6.54	0.71
SN (exch.)	(24, 6)	9.08	0.55	6.40	0.83
MGP	(24, 6)	9.27	0.62	5.83	0.95
CUSP	(24, 6)	9.23	0.57	3.05	0.81
SN (expcov.)	(48, 9)	12.33	0.79	7.59	1.97
SN (fixed)	(48, 9)	12.28	0.75	7.75	1.51
SN (exch.)	(48, 9)	12.29	0.77	6.17	1.60
MGP	(48, 9)	12.39	0.92	6.26	1.34
CUSP	(48, 9)	12.20	0.88	3.05	0.34
SN (expcov.)	(72, 9)	14.08	0.85	7.03	3.04
SN (fixed)	(72, 9)	13.96	1.06	6.98	2.23
SN (exch.)	(72, 9)	14.11	0.88	5.75	1.04
MGP	(72, 9)	14.59	1.09	6.57	1.92
CUSP	(72, 9)	14.19	0.60	3.00	0.53

S4 Further details on applications

S4.1 Simulation experiment

Table 1 of the main manuscript presents the results of the simulation experiments under the settings in which the proportion of the total variation in $\boldsymbol{\Omega} = \boldsymbol{\Delta} + \boldsymbol{\Sigma}$ explained by the common factors was chosen to be $\beta = 0.99$. Tables S1 and S2 present the corresponding results when $\beta = 0.95$ and $\beta = 0.9$, respectively.

In the main manuscript, Section 6.1.3 focused on the results concerning the posterior mean of the geodesic distance $d_G(\boldsymbol{\Omega}, \boldsymbol{\Omega}_0)$. Although of secondary importance, we consider here the corresponding results regarding the posterior mean of the effective number of factors k^* . From Table 1 and Tables S1–S2 it is clear that the results are more mixed than those for $E\{d_G(\boldsymbol{\Omega}, \boldsymbol{\Omega}_0)|\mathbf{y}\}$, with a greater degree of prior sensitivity. In this case, the posteriors under the MGP, CUSP and SN (exch.) priors tend to suggest that k^* takes smaller values than under the SN (expcov.) and SN (fixed) priors. Correspondingly, the posterior mean for k^* tends to be closer to the value used to simulate the data under the former priors when k is smaller (when $(p = 24, k = 6)$) and under the latter priors when k is larger (when $(p = 48, k = 9)$ and $(p = 72, k = 9)$). Interestingly, under the CUSP prior, although the prior mean is $E(k^*) = \alpha = 5$, the posterior seems to stack at $k^* = 3$ irrespective of the value of k used to simulate the data.

S4.2 Co-occurrence of Finnish birds

S4.2.1 Further comments on the model and prior

As remarked in the main manuscript, the meta-covariates and branch lengths are standardised so that all length-scale parameters $\vartheta_1, \dots, \vartheta_5$ operate on a comparable scale. To this end, before calculating the distances $d_{C,ij}$ in the space of meta-covariates, we standardise the single continuous variable, namely the logarithm of body mass $(x_{1,4}, \dots, x_{50,4})$, by dividing the measurements by their standard deviation. In terms of the evolutionary information, the phylogenetic tree in this example is ultrametric and so the root to tip distance is the same for all leaves. As such, before calculating $d_{P,ij}$, the branch lengths are standardised by scaling to make the common root to tip distance one.

In order to facilitate comparison with inferences under the structured increasing shrinkage (SIS) process prior, our prior for the regression coefficients $\mathbf{B}$ determining the area-specific means $\boldsymbol{\mu}_i = \mathbf{B}^T \mathbf{w}_i$ is identical to that chosen in Schiavon et al. [11]. Specifically, structured borrowing of information between species is allowed by taking $\mathbf{B}^T | \mathbf{K} \sim N_{p,c}(\mathbf{X}\mathbf{K}, \mathbf{I}_p, s_\beta^2 \mathbf{I}_c)$ in which $s_\beta^2 = 1$ and $\mathbf{K} = (\kappa_{ij})$ is a $q \times c$ matrix of coefficients. These coefficients are themselves assigned a prior such that $\kappa_{ij} \sim N(0, s_\kappa^2)$ independently in which $s_\kappa^2 = 1$. Finally, following Schiavon et al. the continuous covariates in $\mathbf{W}$ and $\mathbf{X}$ are standardised.

S4.2.2 Posterior summaries

Figure 2 of the main manuscript shows the mean of the posterior for the marginal correlation matrix, $\mathbf{S}_\Omega^{-1} \boldsymbol{\Omega} \mathbf{S}_\Omega^{-1}$, where $\boldsymbol{\Omega} = (\omega_{ij}) = \boldsymbol{\Delta} + \boldsymbol{\Sigma}$ and $\mathbf{S}_\Omega = \text{diag}(\sqrt{\omega_{11}}, \dots, \sqrt{\omega_{pp}})$ under the structured matrix- t prior. The corresponding plot under the structured increasing shrinkage process prior is shown in Figure S2.

S4.2.3 Pseudo marginal likelihood

Denote by $\boldsymbol{\theta}$ the set of all parameters in either of the two model-prior combinations discussed in the main manuscript. Recall that the $n \times p$ binary matrix of occurrences

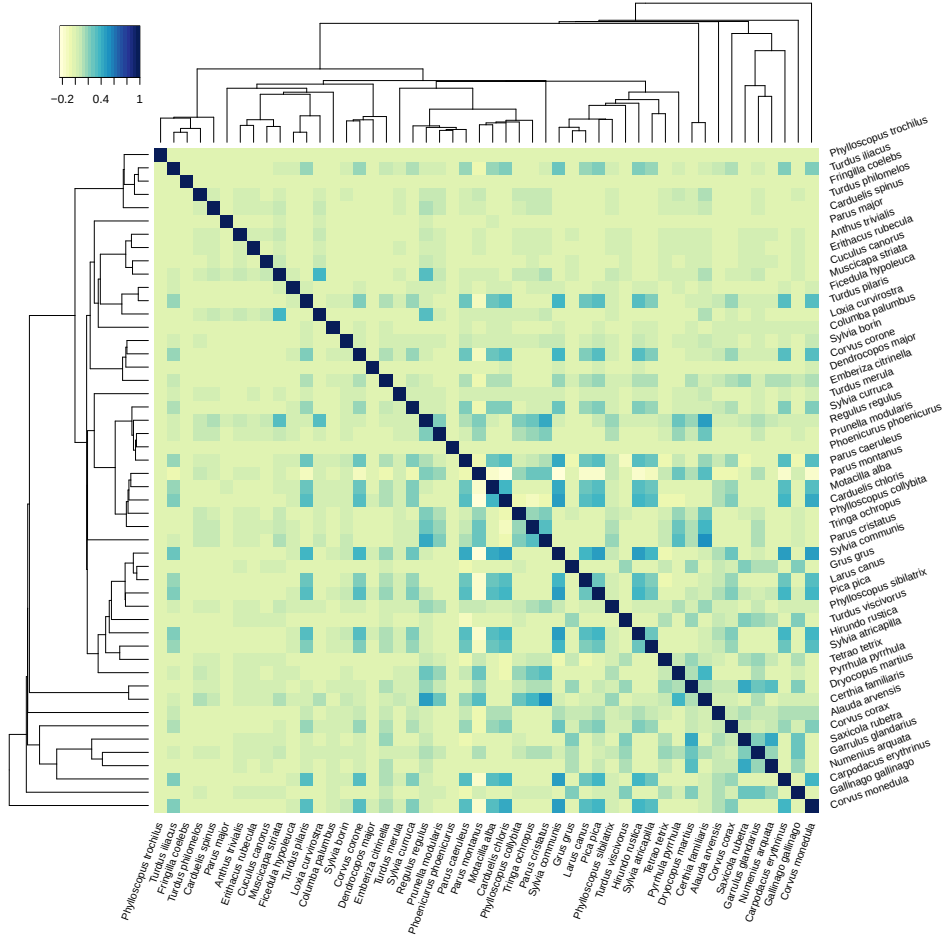


Fig. S2: Mean of the posterior for the marginal correlation matrix under the SIS process prior. The species are ordered according to their phylogenetic tree which is visualised in the margins.

of birds is denoted by $\mathbf{Y}$. Let $\mathbf{y}_i^T$ represent the i^{th} row of $\mathbf{Y}$ and $\mathbf{Y}_{-i}$ represent $\mathbf{Y}$ with the i^{th} row removed. The pseudo marginal likelihood (PML) can then be defined as

$$\text{PML}(\mathbf{Y}) = \prod_{i=1}^n p(\mathbf{y}_i | \mathbf{Y}_{-i}) = \prod_{i=1}^n \int p(\mathbf{y}_i | \boldsymbol{\theta}) \pi(\boldsymbol{\theta} | \mathbf{Y}_{-i}) d\boldsymbol{\theta} \simeq \prod_{i=1}^n \int p(\mathbf{y}_i | \boldsymbol{\theta}) \pi(\boldsymbol{\theta} | \mathbf{Y}) d\boldsymbol{\theta},$$

in which the estimate allows $\text{PML}(\mathbf{Y})$ to be approximated from the output of a single sample, $\boldsymbol{\theta}^{[1]}, \dots, \boldsymbol{\theta}^{[B]}$, from the full posterior $\pi(\boldsymbol{\theta}|\mathbf{Y})$ through

$$\text{PML}(\mathbf{Y}) \simeq \prod_{i=1}^n \left\{ \frac{1}{B} \sum_{b=1}^B \frac{1}{p(\mathbf{y}_i|\boldsymbol{\theta}^{[b]})} \right\}. \quad (\text{S31})$$

Here, each conditional predictive ordinate, $p(\mathbf{y}_i|\mathbf{Y}_{-i})$, has been approximated using a harmonic mean. In practice, each of the terms $p(\mathbf{y}_i|\boldsymbol{\theta}^{[b]})$ in (S31) is a multivariate normal probability which is approximated using the `pmvnorm` function from the `mvtnorm` package in R [12, 13].

S4.2.4 Brier score

The Brier score can readily be calculated using output from the MCMC sampler [14]. For species j and a new sampling location i , the posterior predictive probability that y_{ij} is equal to 1 can be calculated through

$$p_{ij} = \int \Pr(y_{ij} = 1|\boldsymbol{\theta})\pi(\boldsymbol{\theta}|\mathbf{Y}) \simeq \frac{1}{B} \sum_{b=1}^B \Pr(y_{ij} = 1|\boldsymbol{\theta}^{[b]})$$

in which $\boldsymbol{\theta}^{[1]}, \dots, \boldsymbol{\theta}^{[B]}$ are a sample from the posterior $\pi(\boldsymbol{\theta}|\mathbf{Y})$. In order to use all the observations as both test data and training data, we adopt a 4-fold cross-validation approach, randomly assigning the $n = 137$ observations into 4 folds. Denote by $f(k)$ the subset of $\{1, \dots, n\}$ containing observations assigned to fold k and by $\mathbf{Y}_{-f(k)}$ the occurrence matrix with the rows in $f(k)$ removed. We can then calculate

$$p_{ij} \simeq \frac{1}{B} \sum_{b=1}^B \Pr(y_{ij} = 1|\boldsymbol{\theta}^{[b]}), \quad \boldsymbol{\theta}^{[b]} \sim \pi(\boldsymbol{\theta}|\mathbf{Y}_{-f(k)}), \quad b = 1, \dots, B$$

for $i \in f(k)$, $j = 1, \dots, p = 50$ and $k = 1, \dots, 4$. The two model-prior combinations can then be compared based on the mean

$$S = \frac{1}{np} \sum_{i=1}^n \sum_{j=1}^p S(p_{ij}, y_{ij})$$

of the Brier scores, each of which is defined as

$$S(p_{ij}, y_{ij}) = -2(p_{ij} - y_{ij})^2.$$

S4.3 Hourly demand for natural gas

S4.3.1 Model for the time-varying mean

The main factors that are known to influence the mean level of residential demand for gas are the weather along with seasonal and calendar effects [15]. Each term in the

time-varying mean $\boldsymbol{\mu}_t = (\mu_{t1}, \dots, \mu_{t,24})^\top$ can be written as

$$\mu_{ti} = \beta_{1i} + (\beta_{2i} + \beta_{3i}\tilde{w}_t)w_{t2} + \beta_{4i}w_{t4} + \chi_{ti} + \varpi_{ti}$$

where $\tilde{w}_t$ is the temperature for day t and $w_{t2} = \tilde{w}_t - m_{d(t)}$ in which $m_{d(t)}$ is a smoothed annual average for the temperature on day t where $d(t) \in \{1, 2, \dots, 366\}$. We include both the mean-centered temperature, w_{t2} , and its interaction with the raw temperature, $w_{t3} = \tilde{w}_t w_{t2}$, to allow the effect of above or below average temperatures to differ according to absolute weather conditions. The variable w_{t4} is an indicator that is equal to 1 if day t is a public holiday and equal to zero otherwise.

The term χ_{ti} gives a day-of-the-week effect, whilst ϖ_{ti} gives a seasonal or, day-of-the-year, effect. Both terms are composed using Fourier series with

$$\chi_{ti} = \sum_{k=1}^3 \left\{ \beta_{4+2(k-1)+1,i} \cos\left(\frac{2\pi kt}{7}\right) + \beta_{4+2k,i} \sin\left(\frac{2\pi kt}{7}\right) \right\} \quad (\text{S32})$$

and

$$\varpi_{ti} = \sum_{k=1}^K \left\{ \beta_{10+2(k-1)+1,i} \cos\left(\frac{2\pi kt}{365.25}\right) + \beta_{10+2k,i} \sin\left(\frac{2\pi kt}{365.25}\right) \right\}. \quad (\text{S33})$$

The six unconstrained coefficients in (S32) provide fixed effects for each day of the week which, by construction, sum to zero. The advantages of this parameterisation are that the six coefficients are unconstrained and by treating them as exchangeable *a priori*, we can induce a prior for the seven fixed effects which is symmetric with respect to the day of the week. The Fourier series for the seasonal term in (S33) is truncated at $K = 6$ harmonics based on analysis of similar data that suggested the contribution of higher frequency harmonics was negligible [16].

Row i of the $c \times p$ matrix of regression coefficients $\mathbf{B} = (\beta_{ij})$ characterises the effect of the i^{th} covariate on the demand for gas across the $p = 24$ hours of the day. In the prior we assume the coefficients on different rows are independent but allow borrowing of strength within each row by assigning a hierarchical prior. For row i , $i = 1, \dots, c$, this takes the form

$$\beta_{ij} | \mu_{\beta,i}, \sigma_{\beta,i}^2 \sim \text{N}(\mu_{\beta,i}, \sigma_{\beta,i}^2)$$

for $j = 1, \dots, p$ in which $\mu_{\beta,i} \sim \text{N}(0, 7)$ and $1/\sigma_{\beta,i}^2 \sim \gamma(21, 60)$.

S4.3.2 Posterior summaries

The observation equation of the identified model $\mathbf{y}_t = \boldsymbol{\mu}_t + \tilde{\mathbf{\Lambda}}\tilde{\boldsymbol{\eta}}_t + \boldsymbol{\epsilon}_t$ suggests that we can interpret the columns of the identified factor loadings matrix $\tilde{\mathbf{\Lambda}}$ as typical *gas-day-profiles*; on any particular day, $\mathbf{y}_t - \boldsymbol{\mu}_t$ is simply a noisy weighted sum of gas-day-profiles, and these weights evolve over time. Figure S3 shows the posterior mean for the identified factor loadings matrix $\tilde{\mathbf{\Lambda}}$. It appears that many of the columns have

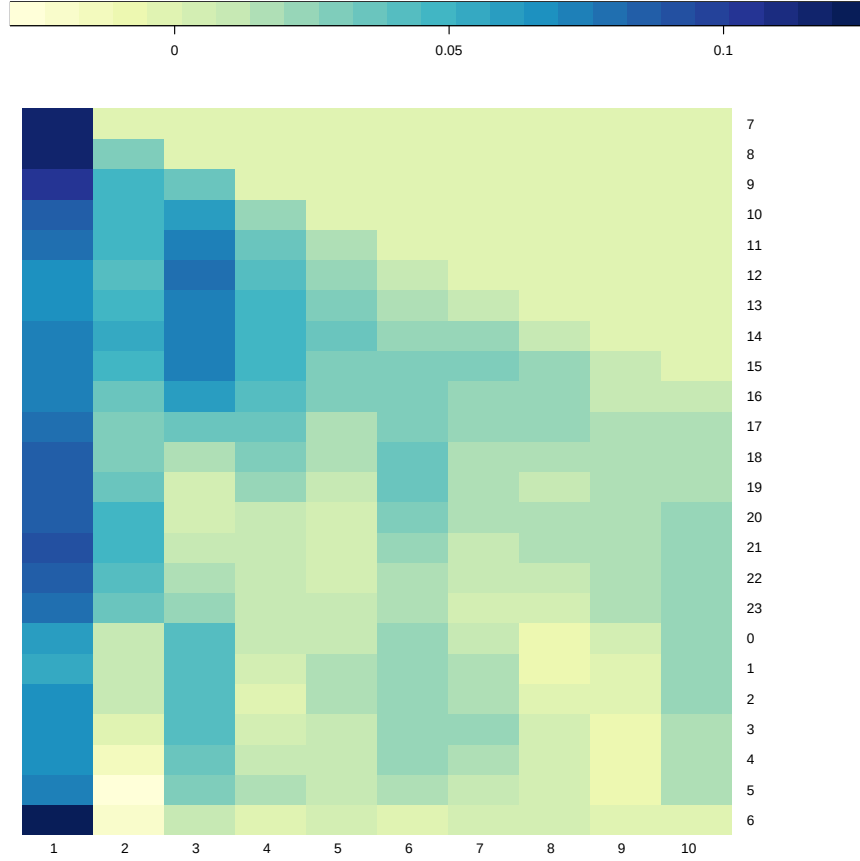


Fig. S3: Mean of the posterior for the identified factor loadings matrix $\tilde{\Lambda}$. The labels on the right vertical axis indicate hour of the day in a 24-hour clock.

a double-hump shape with smaller loadings towards the middle of the gas-day. This is also apparent in Figure S4 which shows a more complete summary of the marginal posterior densities for the loadings of the first three factors, $\tilde{\lambda}_{ii}, \dots, \tilde{\lambda}_{24,i}$ for $i = 1, 2, 3$. Again, this may be due to people switching their heating on twice per day and is therefore consistent with conclusions drawn from the posterior mean of the marginal precision matrix displayed in Figure 3 of the main text.

Figure 4 in the paper showed a plot of the 1- and 24-step ahead posterior predictive distributions for the first 5% of times in the hold-out period. Figure S5 shows an analogous plot for the last 5% of times in the hold-out period and, again, reveals excellent agreement between the mean of the posterior predictive distribution and the test data, as well as narrow credible intervals.

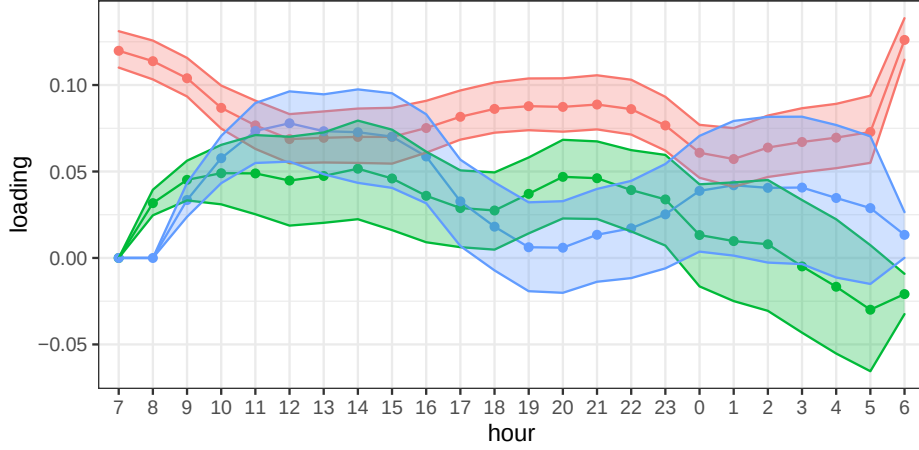


Fig. S4: Summaries of the marginal posterior densities for the non-zero loadings of the first three factors in the identified factor loadings matrix $\tilde{\Lambda}$. The densities for factor 1 (—), factor 2 (—) and factor 3 (—) are summarised through their means and 95% equi-tailed credible intervals.

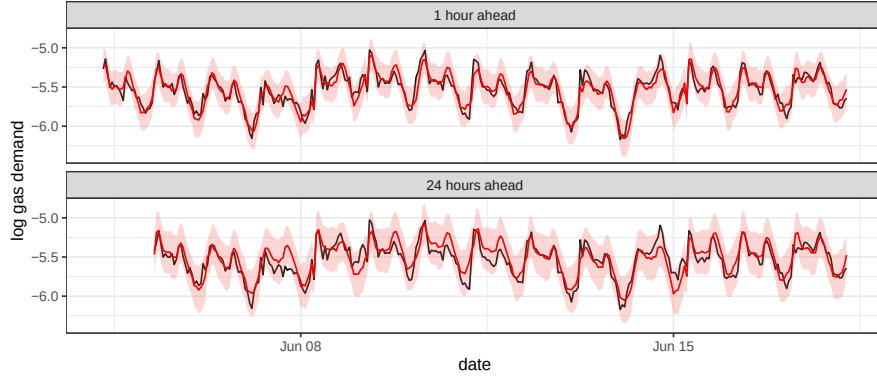


Fig. S5: For the last 5% of times in the hold-out period, means (—) and 95% equi-tailed credible intervals (—) for the one-step ahead and 24-step ahead posterior predictive distributions. The observed data are also shown (—).

S4.3.3 Within-day filtering

For consistency with standard notation [see, for example 17], in the following two sections we denote the observation and state equation as

$$\mathbf{y}_t = \mathbf{F}\boldsymbol{\eta}_t + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}_p(\mathbf{0}, \mathbf{V})$$

and

$$\boldsymbol{\eta}_t = \mathbf{G}\boldsymbol{\eta}_{t-1} + \boldsymbol{\zeta}_t, \quad \boldsymbol{\zeta}_t \sim \mathcal{N}_k(\mathbf{0}, \mathbf{W}).$$

Therefore, for the gas demand application, $\mathbf{y}_t$ represents mean-centered gas demand $\mathbf{y}_t - \boldsymbol{\mu}_t$, $\mathbf{F} = \mathbf{\Lambda}$, $\mathbf{G} = \mathbf{\Gamma}$, $\mathbf{V} = \mathbf{\Sigma}$, $\mathbf{W} = \mathbf{\Pi}$ and $\boldsymbol{\eta}_0 \sim \mathcal{N}_k(\mathbf{0}, \mathbf{I}_k)$.

Let $\mathbf{y}_{s:t}$ denote the collection of vectors $(\mathbf{y}_s, \dots, \mathbf{y}_t)$ and let $\mathbf{y}_{t,g:h}$ denote the sub-vector $(y_{t,g}, \dots, y_{t,h})^\top$. Then at hour $h-1$ of day t , we write the filtering distribution as

$$\boldsymbol{\eta}_t | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:(h-1)} \sim \mathcal{N}_k(\mathbf{m}_t^{(h-1)}, \mathbf{C}_t^{(h-1)})$$

for $h = 2, \dots, p$, noting that when $h = 1$, the corresponding distribution is

$$\boldsymbol{\eta}_{t-1} | \mathbf{y}_{1:(t-1)} \sim \mathcal{N}_k(\mathbf{m}_{t-1}^{(p)}, \mathbf{C}_{t-1}^{(p)})$$

so that $\mathbf{m}_t^{(0)} = \mathbf{m}_{t-1}^{(p)}$ and $\mathbf{C}_t^{(0)} = \mathbf{C}_{t-1}^{(p)}$.

At hour h of day t , we then:

1. Perform a prediction step if $h = 1$ by calculating

$$\begin{aligned} \mathbf{a}_t &= E(\boldsymbol{\eta}_t | \mathbf{y}_{1:(t-1)}) = \mathbf{G}\mathbf{m}_{t-1}^{(p)} \\ \mathbf{R}_t &= \text{Var}(\boldsymbol{\eta}_t | \mathbf{y}_{1:(t-1)}) = \mathbf{G}\mathbf{C}_{t-1}^{(p)}\mathbf{G}^\top + \mathbf{W} \end{aligned}$$

and then

$$\begin{aligned} \mathbf{f}_t^{(0)} &= E(\mathbf{y}_t | \mathbf{y}_{1:(t-1)}) = \mathbf{F}\mathbf{a}_t \\ \mathbf{Q}_t^{(0)} &= \text{Var}(\mathbf{y}_t | \mathbf{y}_{1:(t-1)}) = \mathbf{F}\mathbf{R}_t\mathbf{F}^\top + \mathbf{V} \\ \mathbf{B}_t^{(0)} &= \text{Cov}(\boldsymbol{\eta}_t, \mathbf{y}_t | \mathbf{y}_{1:(t-1)}) = \mathbf{R}_t\mathbf{F}^\top. \end{aligned}$$

We then construct

$$\mathbf{g}_t^{(0)\top} = (\mathbf{f}_t^{(0)\top}, \mathbf{a}_t^\top)$$

and

$$\mathbf{S}_t^{(0)} = \begin{pmatrix} \mathbf{Q}_t^{(0)} & \mathbf{B}_t^{(0)\top} \\ \mathbf{B}_t^{(0)} & \mathbf{R}_t \end{pmatrix}.$$

If $h > 1$, we will have available from the last prediction step $\mathbf{g}_t^{(h-1)}$ and $\mathbf{S}_t^{(h)}$. Note that $\mathbf{g}_t^{(h)} = E\{(\mathbf{y}_{t,(h+1):p}^\top, \boldsymbol{\eta}_t^\top)^\top | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:h}\}$ is a $(k + p - h)$ -vector and $\mathbf{S}_t^{(h)} = \text{Var}\{(\mathbf{y}_{t,(h+1):p}^\top, \boldsymbol{\eta}_t^\top)^\top | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:h}\}$ is a $(k + p - h) \times (k + p - h)$ matrix for $h = 0, \dots, p-1$ which can be partitioned as

$$\mathbf{g}_t^{(h)\top} = (g_{t,1}^{(h)}, \mathbf{g}_{t,-1}^{(h)\top})$$

and

$$\mathbf{S}_t^{(h)} = \begin{pmatrix} S_{t,1,1}^{(h)} & \mathbf{s}_{t,-1,1}^{(h)\top} \\ \mathbf{s}_{t,-1,1}^{(h)} & \mathbf{S}_{t,-1,-1}^{(h)} \end{pmatrix}.$$

2. Perform an update step by using the rules of conditioning for multivariate normal distributions to calculate

$$\begin{aligned} \mathbf{g}_t^{(h)} &= \mathbf{g}_{t,-1}^{(h-1)} + \mathbf{s}_{t,-1,1}^{(h-1)}(y_{t,h} - g_{t,1}^{(h-1)})/S_{t,1,1}^{(h-1)} \\ \mathbf{S}_t^{(h)} &= \mathbf{S}_{t,-1,-1}^{(h-1)} - \mathbf{s}_{t,-1,1}^{(h-1)}\mathbf{s}_{t,-1,1}^{(h-1)\top}/S_{t,1,1}^{(h-1)} \end{aligned}$$

and then

$$\begin{aligned} \mathbf{m}_t^{(h)} &= E(\boldsymbol{\eta}_t | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:h}) = \mathbf{g}_{t,(p-h+1):(k+p-h)}^{(h)} \\ \mathbf{C}_t^{(h)} &= \text{Var}(\boldsymbol{\eta}_t | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:h}) = \mathbf{S}_{t,(p-h+1):(k+p-h),(p-h+1):(k+p-h)}^{(h)}. \end{aligned}$$

If $h < p$ it is also useful for the forecasting recursions (see next section) to note the within-day forecast means and variances

$$\begin{aligned} \mathbf{f}_t^{(h)} &= E(\mathbf{y}_{t,(h+1):p} | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:h}) = \mathbf{g}_{t,1:(p-h)}^{(h)} \\ \mathbf{Q}_t^{(h)} &= \text{Var}(\mathbf{y}_{t,(h+1):p} | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:h}) = \mathbf{S}_{t,1:(p-h),1:(p-h)}^{(h)}. \end{aligned}$$

S4.3.4 Within-day forecasting

Consider generating forecasts using the data up to and including hour h of day t using the output of the filtering algorithm at that time, namely $\mathbf{m}_t^{(h)}$, $\mathbf{C}_t^{(h)}$, $\mathbf{f}_t^{(h)}$ and $\mathbf{Q}_t^{(h)}$. For simplicity in presentation, the algorithm below characterises the predictive distribution for forecasts that are 1 to $dp + h$ hours ahead where $d \geq 0$ can only take integer values and represents the number of days we wish to look into the future.

Denote $\mathbf{f}_t^{(h)}(\ell) = E(\mathbf{y}_{t+\ell} | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:h})$ and $\mathbf{Q}_t^{(h)}(\ell) = \text{Var}(\mathbf{y}_{t+\ell} | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:h})$ and then proceed as follows:

1. Set $\ell = 0$. If $h < p$, we define $\mathbf{f}_t^{(h)}(0) = (\mathbf{y}_{t,1:h}^\top, \mathbf{f}_t^{(h)\top})^\top$ and

$$\mathbf{Q}_t^{(h)}(0) = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{Q}_t^{(h)} \end{pmatrix}$$

since the first h elements of $\mathbf{y}_t$ have already been observed. If $h = p$, we define $\mathbf{f}_t^{(p)}(0) = \mathbf{f}_{t+1}^{(0)} = \mathbf{F}\mathbf{a}_t^{(p)}(0)$ where $\mathbf{a}_t^{(p)}(0) = \mathbf{G}\mathbf{m}_t^{(p)}$ and $\mathbf{Q}_t^{(p)}(0) = \mathbf{Q}_{t+1}^{(0)} = \mathbf{F}\mathbf{R}_t^{(p)}(0)\mathbf{F}^\top + \mathbf{V}$ where $\mathbf{R}_t^{(p)}(0) = \mathbf{G}\mathbf{C}_t^{(p)}\mathbf{G}^\top + \mathbf{W}$.

2. Then if $d > 0$:

- (a) If $h < p$, we initialise with $\mathbf{a}_t^{(h)}(0) = \mathbf{m}_t^{(h)}$ and $\mathbf{R}_t^{(h)}(0) = \mathbf{C}_t^{(h)}$. If $h = p$, we have already calculated $\mathbf{a}_t^{(p)}(0)$ and $\mathbf{R}_t^{(p)}(0)$ in (1).
- (b) For $\ell = 1, \dots, d$, we can then recursively calculate

$$\begin{aligned} \mathbf{a}_t^{(h)}(\ell) &= E(\boldsymbol{\eta}_{t+\ell} | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:h}) = \mathbf{G}\mathbf{a}_t^{(h)}(\ell-1) \\ \mathbf{R}_t^{(h)}(\ell) &= \text{Var}(\boldsymbol{\eta}_{t+\ell} | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:h}) = \mathbf{G}\mathbf{R}_t^{(h)}(\ell-1)\mathbf{G}^\top + \mathbf{W} \end{aligned}$$

and

$$\begin{aligned}\mathbf{f}_t^{(h)}(\ell) &= E(\mathbf{y}_{t+\ell} | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:h}) = \mathbf{F} \mathbf{a}_t^{(h)}(\ell) \\ \mathbf{Q}_t^{(h)}(\ell) &= \text{Var}(\mathbf{y}_{t+\ell} | \mathbf{y}_{1:(t-1)}, \mathbf{y}_{t,1:h}) = \mathbf{F} \mathbf{R}_t^{(h)}(\ell) \mathbf{F}^T + \mathbf{W}.\end{aligned}$$

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